Shor's Algorithm

Number theory, order finding, phase estimation, implementation and cryptography

ANSWER BOOK

This book accompanies the separate *Question Paper* (80 questions, 3 hours, 240 marks). It contains

1. the **answer key**, section by section, with the topic tested by every question (page 2),
2. a **topic-wise map** grouping the questions by theme and a **quick-marking grid** (page 3), and
3. the **Answer Bank**: every question restated with the correct option marked ✓ and a complete solution with derivations (from page 4).

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks:	+3 if ONLY the correct option is chosen;
Zero marks:	0 if none of the options is chosen (question unanswered);
Negative marks:	−1 in all other cases (wrong option, or more than one option chosen).

How to evaluate

For each section count N_c = number of correct answers and N_w = number of questions answered wrongly (or with more than one option marked). Section marks $= 3N_c - N_w$; unanswered questions score 0. The total is out of 240. Every numerical answer was re-computed independently (exact integer and `Fraction` arithmetic, NumPy phase-estimation formulas, checks against the code and recorded outputs of `Shor.ipynb`, and CUDA-Q state-vector and density-matrix simulations with the notebook's circuit builder) by `verify_answers.py` in the source folder.

Conventions (identical to the Question Paper)

- $a \bmod N$ is the remainder in $\{0, \ldots, N-1\}$; the **order** $r = \mathrm{ord}_N(a)$ is the least $r > 0$ with $a^r \equiv 1 \pmod N$ (defined for $\gcd(a, N) = 1$); φ is Euler's totient, λ the Carmichael function. "Factors" means non-trivial factors ($1 < p < N$).
- **Reduction:** if r is even and $a^{r/2} \not\equiv -1 \pmod N$, then $\gcd(a^{r/2} - 1, N)$ and $\gcd(a^{r/2} + 1, N)$ are non-trivial factors; otherwise the attempt fails.
- **Circuit:** t counting qubits (Hadamards), work register in $|1\rangle$, controlled $U_{a^{2^j} \bmod N}$ with $U_c\,|y\rangle = |cy \bmod N\rangle$ for $y < N$, inverse QFT, measurement of the counting register giving $y \in \{0, \ldots, 2^t - 1\}$ (most significant bit first). **Post-processing:** `Fraction(y, 2**t).limit_denominator(N)` gives s/r'; the order is r' or its smallest multiple kr' with $a^{kr'} \equiv 1$. $y = 0$ gives no candidate.
- **Notebook** (`Shor.ipynb`, defaults): $N = 15$, $a = 7$, $n =$ `N.bit_length()` work qubits, $t = 2n$ counting qubits, multipliers as exact permutation matrices (`cudaq.register_operation`), 1000 shots. **Previously recorded run**: CUDA-Q 0.16.0, CPU target `qpp-cpu`, no GPU; 12 qubits, 51 gates (plus 8 measurements), outcomes $0, 64, 128, 192$; 0.766 of the shots gave $15 = 3 \times 5$. No physical quantum processor was used.
- State-vector memory: 16 bytes per amplitude (2^q amplitudes for q qubits). 1 MB $= 10^6$ bytes, 1 GB $= 10^9$ bytes.

Answer key

Answer distribution: (A) 20, (B) 20, (C) 20, (D) 20.

Section A — Mathematical and Number Theory Foundations

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
1	**A**	Euclidean algorithm	11	**A**	Failure: $a^{r/2} \equiv -1$
2	**A**	Modular inverse	12	**C**	Fermat's little theorem
3	**D**	Euler's totient $\varphi(pq)$	13	**D**	Trial-division bound
4	**B**	Multiplicative order	14	**C**	Success bound $1 - 2^{-(k-1)}$
5	**B**	Repeated squaring	15	**C**	φ, order, λ, inverse
6	**D**	Square-and-multiply cost	16	**D**	Classical factoring complexity
7	**B**	Orders divide $\lambda(N)$	17	**A**	Non-trivial square root $\Rightarrow$ factor
8	**A**	Lucky gcd probability	18	**C**	Euler's theorem vs order
9	**D**	Chinese remainder theorem	19	**B**	Order of 3 mod 91
10	**B**	Square roots of 1 mod 15	20	**C**	Factors of 91 by gcd

Section B — Quantum Computing Foundations

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
21	**B**	Uniform superposition	31	**B**	CNOT cost of a Fredkin gate
22	**D**	Measuring the work register	32	**C**	Implicit measurement
23	**C**	Unitarity of U_a	33	**A**	Notebook qubit count
24	**D**	Eigenvalues of U_a	34	**D**	State-vector memory
25	**B**	$\|1\rangle$ as sum of eigenvectors	35	**B**	Roles of circuit blocks
26	**D**	Phase kickback	36	**B**	Phase gates $R_1(\lambda)$
27	**C**	Phases of U^{2^j}	37	**C**	Kickback and controls
28	**A**	Hadamard transform signs	38	**D**	Parallelism vs interference
29	**C**	Two-path interference	39	**A**	QPE: exact phase
30	**A**	Reversible modular exponentiation	40	**A**	QPE: inexact phase

Section C — Shor's Algorithm Fundamentals

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
41	**D**	Steps of Shor's algorithm	51	**D**	Exact success, notebook default
42	**B**	$N = 15$, $a = 2$	52	**A**	Probability $\varphi(r)/r$
43	**C**	Failure for $a = N - 1$	53	**A**	Role of the QFT in Shor
44	**A**	Odd order: retry	54	**C**	Skipped identity multipliers
45	**C**	Lucky base	55	**B**	Functions of `Shor.ipynb`
46	**A**	Register sizes for $N = 33$	56	**D**	Outcomes to fractions
47	**C**	$N = 15$, $a = 11$ outcomes	57	**D**	Prime powers
48	**D**	Continued fractions, $N = 33$	58	**B**	Uninformative outcomes
49	**B**	Multiples test	59	**A**	$N = 39$: order and factors
50	**B**	Condition $2^t \geq N^2$	60	**C**	$N = 39$: continued fraction

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Section D — Advanced Shor's Algorithm and Implementation

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
61	**B**	Beauregard qubit count	71	**D**	Memory for $N = 1023$
62	**C**	Published RSA-2048 estimates	72	**A**	Toy RSA
63	**C**	$O(n^3)$ scaling	73	**A**	Post-quantum standards
64	**D**	Semiclassical QFT	74	**C**	Impact on RSA, ECC, AES
65	**A**	Approximate QFT	75	**D**	Fault-tolerance terms
66	**B**	Controlled phases in the QFT	76	**C**	Qubit counts of constructions
67	**A**	Recorded gate counts	77	**B**	Harvest now, decrypt later
68	**D**	No hardware execution	78	**A**	Qubits and repeated squaring
69	**C**	CNOTs of a controlled multiplier	79	**B**	Error-free probability vs success
70	**B**	Random baseline	80	**D**	Interpreting noisy results

Topic-wise map of the questions

Every question appears under at least one theme; several appear under two.

Theme	Question numbers
Modular arithmetic, gcd, inverses and totients	1, 2, 3, 9, 12, 15, 72
Multiplicative order and repeated squaring	4, 5, 6, 7, 18, 19, 27, 78
Reduction of factoring to order finding	8, 10, 11, 14, 17, 20, 41, 42, 43, 44, 45, 57, 59
Classical factoring and complexity	13, 16, 63
Superposition, entanglement and measurement	21, 22, 30, 32, 38
Unitary U_a, eigenvectors and phase kickback	23, 24, 25, 26, 37
Gates and interference	28, 29, 31, 36, 66, 69
Phase estimation and the QFT	35, 39, 40, 53, 64, 65, 66
Outcomes, continued fractions and post-processing	47, 48, 49, 50, 51, 52, 56, 58, 60
Register sizes, memory and simulation limits	33, 34, 46, 50, 71, 78
`Shor.ipynb`: code and recorded results	51, 54, 55, 67, 68
Gate-level circuits and noise	31, 69, 70, 79, 80
Resource estimates and fault tolerance	61, 62, 64, 75, 76
RSA, ECC and post-quantum cryptography	72, 73, 74, 77

Quick-marking grid

1 **A**	2 **A**	3 **D**	4 **B**	5 **B**	6 **D**	7 **B**	8 **A**	9 **D**	10 **B**
11 **A**	12 **C**	13 **D**	14 **C**	15 **C**	16 **D**	17 **A**	18 **C**	19 **B**	20 **C**
21 **B**	22 **D**	23 **C**	24 **D**	25 **B**	26 **D**	27 **C**	28 **A**	29 **C**	30 **A**
31 **B**	32 **C**	33 **A**	34 **D**	35 **B**	36 **B**	37 **C**	38 **D**	39 **A**	40 **A**
41 **D**	42 **B**	43 **C**	44 **A**	45 **C**	46 **A**	47 **C**	48 **D**	49 **B**	50 **B**
51 **D**	52 **A**	53 **A**	54 **C**	55 **B**	56 **D**	57 **D**	58 **B**	59 **A**	60 **C**
61 **B**	62 **C**	63 **C**	64 **D**	65 **A**	66 **B**	67 **A**	68 **D**	69 **C**	70 **B**
71 **D**	72 **A**	73 **A**	74 **C**	75 **D**	76 **C**	77 **B**	78 **A**	79 **B**	80 **D**

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Answer Bank — questions with detailed solutions

SECTION A · Mathematical and Number Theory Foundations Q.1–20

Q.1 Single-correct MCQ · Euclidean algorithm

Using the Euclidean algorithm, $\gcd(1071, 462)$ equals

(A) ✓ 21 **(B)** 7 **(C)** 147 **(D)** 42

Answer: (A) Detailed solution

Correct answer. (A) 21

Concepts required. The Euclidean algorithm replaces (a, b) by $(b, a \bmod b)$ until the remainder is 0; the last non-zero remainder is the gcd.

Relevant formulae. $\gcd(a, b) = \gcd(b, a \bmod b)$: $1071 = 2 \cdot 462 + 147$, $462 = 3 \cdot 147 + 21$, $147 = 7 \cdot 21 + 0$.

Step-by-step explanation. $1071 = 2 \cdot 462 + 147$; $462 = 3 \cdot 147 + 21$; $147 = 7 \cdot 21 + 0$. The last non-zero remainder is 21. *Check*: $1071 = 3^2 \cdot 7 \cdot 17$ and $462 = 2 \cdot 3 \cdot 7 \cdot 11$, common part $3 \cdot 7 = 21$. The algorithm needs $O(\log N)$ division steps, which is why the gcd steps of Shor's algorithm are classically cheap.

Why the other options are incorrect.

(B) 7 divides both numbers, but so does $21 = 3 \cdot 7$; 7 is a common divisor, not the greatest one.

(C) 147 is the first remainder of the algorithm, not its last non-zero remainder; $147 \nmid 462$.

(D) 42 does not divide 1071 (odd), so it cannot be a common divisor at all.

Key takeaway. The final gcd step of Shor's algorithm is this fast classical algorithm: $O(\log N)$ division steps.

Q.2 Single-correct MCQ · Modular inverse

The multiplicative inverse of 3 modulo 20 (the b with $3b \equiv 1 \pmod{20}$, $0 < b < 20$) is

(A) ✓ 7 **(B)** 13 **(C)** 17 **(D)** 3

Answer: (A) Detailed solution

Correct answer. (A) 7

Concepts required. An inverse modulo m exists iff $\gcd(b, m) = 1$; it is found by the extended Euclidean algorithm or by testing.

Relevant formulae. $3b \equiv 1 \pmod{20}$: $3 \cdot 7 = 21 = 20 + 1$.

Step-by-step explanation. Extended Euclid: $20 = 6 \cdot 3 + 2$, $3 = 1 \cdot 2 + 1$, so $1 = 3 - 2 = 3 - (20 - 6 \cdot 3) = 7 \cdot 3 - 20$. Hence $3 \cdot 7 = 21 \equiv 1 \pmod{20}$, $b = 7$. The inverse exists because $\gcd(3, 20) = 1$; the same condition $\gcd(a, N) = 1$ makes multiplication by a modulo N reversible.

Why the other options are incorrect.

(B) $3 \cdot 13 = 39 \equiv 19 \equiv -1 \pmod{20}$: 13 is the inverse of -3, not of 3.

(C) $3 \cdot 17 = 51 \equiv 11 \pmod{20}$, which is not 1.

(D) 3 is the number itself; $3 \cdot 3 = 9 \not\equiv 1 \pmod{20}$.

Key takeaway. Invertibility of a modulo N is exactly what makes the multiplier U_a a permutation (unitary).

Q.3 Single-correct MCQ · Euler's totient $\varphi(pq)$

Euler's totient $\varphi(91)$ equals

(A) 84 **(B)** 90 **(C)** 78 **(D)** ✓ 72

Answer: (D) Detailed solution

Correct answer. (D) 72
Concepts required. Euler's totient counts the residues coprime to N; it is multiplicative for coprime factors.
Relevant formulae. $\varphi(pq) = (p-1)(q-1)$ for distinct primes: $\varphi(7 \cdot 13) = 6 \cdot 12 = 72$.
Step-by-step explanation. $91 = 7{\cdot}13$ with distinct primes, so $\varphi(91) = (7-1)(13-1) = 6{\cdot}12 = 72$: the number of $1 \le a \le 91$ coprime to 91. $90 = N - 1$ would hold only for a prime N.
Why the other options are incorrect.
(A) $84 = 91 - 7$ removes only the multiples of 13 (7 of them) and forgets the multiples of 7.
(B) $90 = 91 - 1$ would be φ of a prime; $91 = 7 \cdot 13$ is composite.
(C) $78 = 91 - 13$ removes only the multiples of 7 and forgets the multiples of 13.
Key takeaway. Knowing $\varphi(N)$ is equivalent to knowing the factors of an RSA modulus.

Q.4 Single-correct MCQ · Multiplicative order

The multiplicative order of 2 modulo 11 is

(A) 11 **(B)** ✓ 10 **(C)** 2 **(D)** 5

Answer: (B) Detailed solution

Correct answer. (B) 10
Concepts required. The order is the least $r > 0$ with $a^r \equiv 1$; it divides $\varphi(N)$ (Lagrange).
Relevant formulae. Powers of 2 mod 11: $2, 4, 8, 5, 10, 9, 7, 3, 6, 1$ — the first 1 is at $r = 10$.
Step-by-step explanation. Powers of 2 mod 11: $2, 4, 8, 5, 10, 9, 7, 3, 6, 1$ — the first 1 appears at exponent 10. The order divides $\varphi(11) = 10$, so the only candidates are $1, 2, 5, 10$: $2^2 = 4$, $2^5 = 32 \equiv 10 \equiv -1$, so the order is 10 (2 is a primitive root mod 11).
Why the other options are incorrect.
(A) 11 is the modulus; orders modulo a prime p divide $p - 1 = 10$, so 11 is impossible.
(C) $2^2 = 4 \neq 1$; 2 is the base, not its order.
(D) $2^5 = 32 \equiv 10 \equiv -1 \pmod{11}$, not 1; 5 divides 10 but is not the order.
Key takeaway. Order finding is the hard problem; for tiny N we can list powers, for RSA sizes we cannot.

Q.5 Single-correct MCQ · Repeated squaring

Using repeated squaring, $3^{13} \bmod 17$ equals

(A) 5 **(B)** ✓ 12 **(C)** 3 **(D)** 16

Answer: (B) Detailed solution

Correct answer. (B) 12
Concepts required. Repeated squaring computes a^{2^k} by squaring and multiplies the powers selected by the binary digits of the exponent.
Relevant formulae. $13 = 8+4+1$; $3^2 = 9$, $3^4 = 81 \equiv 13$, $3^8 \equiv 169 \equiv 16$; $3^{13} \equiv 16{\cdot}13{\cdot}3 = 624 \equiv 12 \pmod{17}$.
Step-by-step explanation. $3^1 = 3$, $3^2 = 9$, $3^4 = 81 \equiv 13$, $3^8 = 169 \equiv 16 \pmod{17}$. With

$13 = 8 + 4 + 1$: $3^{13} \equiv 16 \cdot 13 \cdot 3 = 624 = 36 \cdot 17 + 12 \equiv 12$. *Check*: $3^{16} \equiv 1$ (Fermat), so $3^{13} \equiv 3^{-3} = 27^{-1} \equiv 10^{-1} \equiv 12$ since $10 \cdot 12 = 120 = 7 \cdot 17 + 1$.

Why the other options are incorrect.

(A) $5 \equiv 3^5 \pmod{17}$ ($243 = 14 \cdot 17 + 5$); it uses the wrong exponent.

(C) $3 = 3^1$ is the starting value; it would require $3^{12} \equiv 1$, but the order of 3 modulo 17 is 16.

(D) $16 \equiv 3^8$ is only one of the three factors needed for $13 = 8 + 4 + 1$.

Key takeaway. Repeated squaring is the classical model for the controlled-U^{2^j} structure of the quantum circuit.

Q.6 SINGLE-CORRECT MCQ · Square-and-multiply cost

Square-and-multiply computes $a^{1000} \bmod N$ from the binary expansion $1000 = 1111101000_2$. The number of modular multiplications (squarings plus extra multiplications) is

(A) 10 **(B)** 16 **(C)** 999 **(D)** ✓ 14

Answer: (D) Detailed solution

Correct answer. (D) 14

Concepts required. Square-and-multiply performs one squaring per bit after the leading one and one extra multiplication per further 1-bit.

Relevant formulae. $1000 = 1111101000_2$: 10 bits ⇒ 9 squarings; six 1-bits ⇒ 5 multiplications; total 14.

Step-by-step explanation. Left-to-right: start with a (leading bit), then for each of the remaining 9 bits square once (9 squarings) and multiply by a when the bit is 1. The bits after the leading one are 111101000: five 1s, so 5 extra multiplications. Total $9 + 5 = 14$. Naively 999 multiplications would be needed. This is the classical idea behind computing $a^{2^j} \bmod N$ in Shor's circuit.

Why the other options are incorrect.

(A) 10 is the number of bits; it ignores the extra multiplications for the 1-bits (and overcounts the squarings by one).

(B) 16 counts a squaring and a multiplication for each of the six 1-bits plus extras; only $9 + 5$ operations are needed.

(C) 999 multiplications is the naive method $a \cdot a \cdots a$, which is exponential in the bit length.

Key takeaway. Modular exponentiation is polynomial: $O(n)$ multiplications of n-bit numbers, $O(n^3)$ bit operations.

Q.7 SINGLE-CORRECT MCQ · Orders divide $\lambda(N)$

For $N = 35$, which of the following can *not* be the multiplicative order of any a with $\gcd(a, 35) = 1$?

(A) 6 **(B)** ✓ 5 **(C)** 4 **(D)** 12

Answer: (B) Detailed solution

Correct answer. (B) 5

Concepts required. Every order divides the Carmichael function $\lambda(N)$, the exponent of the group $\mathbb{Z}_N^*$.

Relevant formulae. $\lambda(35) = \operatorname{lcm}(\lambda(5), \lambda(7)) = \operatorname{lcm}(4, 6) = 12$; possible orders are the divisors of 12: $1, 2, 3, 4, 6, 12$.

Step-by-step explanation. By the Chinese remainder theorem the order modulo 35 is $\operatorname{lcm}(r_5, r_7)$ with $r_5 \mid 4$ and $r_7 \mid 6$, so every order divides the Carmichael value $\lambda(35) = \operatorname{lcm}(4, 6) = 12$ (and $\varphi(35) = 24$). Orders 12 (e.g. $a = 2$), 4 (e.g. $a = 8$: $8^2 = 64 \equiv 29$, $8^4 \equiv 1$) and 6 (e.g. $a = 4$:

$4^3 = 64 \equiv 29$, $4^6 \equiv 1$) occur; 5 divides neither 12 nor 24.

Why the other options are incorrect.

(A) 6 divides 12 and is attained, e.g. by $a = 4$ ($4^3 = 64 \equiv 29 \not\equiv 1$, $4^6 \equiv 1$, $4^2 = 16$).

(C) 4 divides 12 and is attained, e.g. by $a = 8$ ($8^2 = 64 \equiv 29$, $8^4 \equiv 29^2 = 841 \equiv 1$).

(D) $12 = \lambda(35)$ is attained, e.g. by $a = 3$ ($3^{12} \equiv 1$ and no smaller power is 1).

Key takeaway. Orders divide $\lambda(N)$, which divides $\varphi(N)$; 5 does not divide 12.

Q.8 Single-correct MCQ · Lucky gcd probability

For $N = 15$, a base a is drawn uniformly from $\{2, 3, \ldots, 13\}$. The probability that $\gcd(a, 15) > 1$, so that a factor is found classically without any quantum step, is

(A) ✓ 1/2 **(B)** 8/15 **(C)** 7/15 **(D)** 1/3

Answer: (A) Detailed solution

Correct answer. (A) 1/2

Concepts required. Before any quantum step, Shor's algorithm computes $\gcd(a, N)$; if it exceeds 1 a factor is found classically.

Relevant formulae. In $\{2, \ldots, 13\}$ (12 values) the bases sharing a factor with 15 are $3, 5, 6, 9, 10, 12$: $6/12 = 1/2$.

Step-by-step explanation. The multiples of 3 or 5 in $\{2, \ldots, 13\}$ are $3, 5, 6, 9, 10, 12$: six of the twelve values, probability $\frac{1}{2}$. (For RSA-sized $N = pq$ this probability is about $\frac{1}{p} + \frac{1}{q}$, astronomically small — the lucky gcd is only an artefact of tiny examples.)

Why the other options are incorrect.

(B) $8/15 = \varphi(15)/15$ is the fraction of coprime residues in $\{0, \ldots, 14\}$, the complement of what is asked, on the wrong set.

(C) 7/15 divides by all 15 residues and counts 0; the bases are drawn from 12 values, 6 of which are lucky.

(D) 1/3 counts only the multiples of 3 $(3, 6, 9, 12)$ or of 5 separately; both kinds must be included.

Key takeaway. For tiny N lucky bases are common; for RSA moduli the probability is negligible ($\approx 2/\sqrt{N}$).

Q.9 Single-correct MCQ · Chinese remainder theorem

The unique $x \in \{0, \ldots, 14\}$ with $x \equiv 2 \pmod 3$ and $x \equiv 3 \pmod 5$ is

(A) 11 **(B)** 2 **(C)** 13 **(D)** ✓ 8

Answer: (D) Detailed solution

Correct answer. (D) 8

Concepts required. The Chinese remainder theorem: for coprime moduli, the system has a unique solution modulo their product.

Relevant formulae. $x \equiv 3 \pmod 5$: $x \in \{3, 8, 13\}$; of these only $8 \equiv 2 \pmod 3$.

Step-by-step explanation. Candidates with $x \equiv 3 \pmod 5$: $3, 8, 13$; their residues mod 3 are $0, 2, 1$. So $x = 8$. The Chinese remainder theorem guarantees one solution modulo 15; it is the reason why orders modulo $N = pq$ are least common multiples of the orders modulo p and q.

Why the other options are incorrect.

(A) $11 \equiv 2 \pmod 3$ but $11 \equiv 1 \pmod 5$.

(B) $2 \equiv 2 \pmod 3$ but $2 \not\equiv 3 \pmod 5$.

(C) $13 \equiv 3 \pmod 5$ but $13 \equiv 1 \pmod 3$, not 2.

Key takeaway. CRT explains the four square roots of 1 modulo pq: ± 1 modulo each prime independently.

Q.10 Single-correct MCQ · Square roots of 1 mod 15

The complete set of solutions of $x^2 \equiv 1 \pmod{15}$ with $0 \leq x < 15$ is

(A) $\{1, 11\}$ **(B)** ✓ $\{1, 4, 11, 14\}$ **(C)** $\{1, 14\}$ **(D)** $\{1, 4, 14\}$

Answer: (B) Detailed solution

Correct answer. **(B)** $\{1, 4, 11, 14\}$
Concepts required. Modulo $N = pq$, $x^2 \equiv 1$ iff $x \equiv \pm 1$ modulo both p and q (CRT), giving four solutions.
Relevant formulae. $1^2 = 1$, $4^2 = 16 \equiv 1$, $11^2 = 121 \equiv 1$, $14^2 = 196 \equiv 1 \pmod{15}$.
Step-by-step explanation. By CRT, $x^2 \equiv 1 \bmod 15$ iff $x \equiv \pm 1 \pmod 3$ and $x \equiv \pm 1 \pmod 5$: four combinations, $x = 1, 4, 11, 14$ ($4^2 = 16$, $11^2 = 121 = 8 \cdot 15 + 1$). The non-trivial roots 4 and 11 give factors: $\gcd(4 - 1, 15) = 3$, $\gcd(4 + 1, 15) = 5$. Shor's algorithm finds such a root as $x = a^{r/2}$.
Why the other options are incorrect.

(A) $\{1, 11\}$ misses 4 and 14; $14 \equiv -1$ is always a root.

(C) $\{1, 14\}$ are only the trivial roots ± 1; the non-trivial ones 4 and 11 are what Shor's algorithm exploits.

(D) $\{1, 4, 14\}$ misses $11 = 15 - 4$; roots come in pairs $\pm x$.

Key takeaway. A non-trivial square root of 1 immediately factors N via $\gcd(x \pm 1, N)$.

Q.11 Single-correct MCQ · Failure: $a^{r/2} \equiv -1$

For $N = 21$ and $a = 5$, the order r and the outcome of Shor's classical reduction are

(A) ✓ $r = 6$; the reduction fails because $5^3 \equiv -1 \pmod{21}$

(B) $r = 3$; it fails because r is odd

(C) $r = 12$; it gives the factors 3 and 7

(D) $r = 6$; it gives the factors 3 and 7

Answer: (A) Detailed solution

Correct answer. **(A)** $r = 6$; the reduction fails because $5^3 \equiv -1 \pmod{21}$
Concepts required. The reduction needs r even and $a^{r/2} \not\equiv -1$; otherwise the attempt fails and a new base is chosen.
Relevant formulae. Powers of 5 mod 21: $5, 4, 20, 16, 17, 1 \Rightarrow r = 6$; $5^3 \equiv 20 \equiv -1$.
Step-by-step explanation. $5^1 = 5$, $5^2 = 25 \equiv 4$, $5^3 \equiv 20 \equiv -1$, $5^6 \equiv 1$: $r = 6$ (even). But $a^{r/2} = 5^3 \equiv -1$, so $\gcd(5^3 + 1, 21) = \gcd(21, 21) = 21$ and $\gcd(5^3 - 1, 21) = \gcd(19, 21) = 1$: only trivial factors. The algorithm must restart with another a.
Why the other options are incorrect.

(B) $5^3 \equiv 20 \neq 1$, so 3 is not the order; the order is 6, which is even.

(C) 12 is a multiple of the order, not the order; $5^6 \equiv 1$ already.

(D) With $5^3 \equiv -1$ the gcds are $\gcd(19, 21) = 1$ and $\gcd(21, 21) = 21$: trivial, so no factors are obtained.

Key takeaway. Correct order finding can still give a failed attempt; report it as a failure and retry.

Q.12 SINGLE-CORRECT MCQ · Fermat's little theorem

$2^{100} \bmod 13$ equals

(A) 9 **(B)** 1 **(C)** ✓ 3 **(D)** 12

Answer: (C) Detailed solution

Correct answer. (C) 3
Concepts required. Fermat's little theorem: $a^{p-1} \equiv 1 \pmod p$ for prime $p \nmid a$; reduce the exponent modulo the order.
Relevant formulae. $2^{12} \equiv 1 \pmod{13}$, $100 = 8 \cdot 12 + 4$, so $2^{100} \equiv 2^4 = 16 \equiv 3$.
Step-by-step explanation. Fermat: $2^{12} \equiv 1 \pmod{13}$. $100 = 8 \cdot 12 + 4$, so $2^{100} \equiv 2^4 = 16 \equiv 3$. Reducing the exponent modulo the order (or φ) is what makes $a^x \bmod N$ periodic in x.
Why the other options are incorrect.

(A) $9 \equiv 2^8 \pmod{13}$ ($256 = 19 \cdot 13 + 9$): it reduces the exponent incorrectly; $100 \bmod 12 = 4$, not 8.

(B) 1 would require $12 \mid 100$; $100 = 8 \cdot 12 + 4$.

(D) $12 \equiv -1$ is 2^6; the exponent 100 reduces to 4, not 6.

Key takeaway. Exponents can be reduced modulo the order — which is exactly why the order is so useful.

Q.13 SINGLE-CORRECT MCQ · Trial-division bound

Trial division factors $N = 10403$ by testing divisors $d \geq 2$ in increasing order. The largest d that may have to be tested is

(A) 5201 **(B)** 102 **(C)** 103 **(D)** ✓ 101

Answer: (D) Detailed solution

Correct answer. (D) 101
Concepts required. A composite N has a prime factor $\leq \sqrt{N}$, so trial division stops at $\lfloor \sqrt{N} \rfloor$.
Relevant formulae. $\sqrt{10403} = 101.99$; $10403 = 101 \cdot 103$, so $d = 101$ is reached and divides.
Step-by-step explanation. If N is composite it has a factor $\leq \sqrt{N}$. $101^2 = 10201 \leq 10403 < 102^2 = 10404$, so testing up to $\lfloor \sqrt{N} \rfloor = 101$ suffices — and $10403 = 101 \times 103$ is found at $d = 101$. For an n-bit N this is about $2^{n/2}$ divisions: exponential in n.
Why the other options are incorrect.

(A) $5201 \approx N/2$ is a far too pessimistic bound; divisors above $\sqrt{N}$ pair with ones below it.

(B) $102 > \sqrt{10403}$ is never needed (and is even); the search stops at 101.

(C) 103 is the larger factor; it is found as $N/101$, never by testing.

Key takeaway. Trial division costs $O(\sqrt{N}) = O(2^{n/2})$ — exponential in the bit length.

Q.14 SINGLE-CORRECT MCQ · Success bound $1 - 2^{-(k-1)}$

For an odd N with k distinct prime factors (not a prime power) and a uniformly random among the residues coprime to N, the probability that r is even and $a^{r/2} \not\equiv -1 \pmod{N}$ is at least $1 - 2^{-(k-1)}$. For $N = 105 = 3 \cdot 5 \cdot 7$ this lower bound is

(A) 1/8 **(B)** 1/2 **(C)** ✓ 3/4 **(D)** 7/8

Answer: (C) Detailed solution

Correct answer. (C) 3/4
Concepts required. For N with k distinct odd prime factors, a random coprime base works with probability at least $1 - 2^{-(k-1)}$.
Relevant formulae. $k = 3$: $1 - 2^{-2} = 3/4$.
Step-by-step explanation. $k = 3$: $1 - 2^{-2} = \frac{3}{4}$. For RSA moduli ($k = 2$) the bound is $\frac{1}{2}$, so on average at most two bases are needed. This bound comes from the CRT: the reduction fails only if the 2-adic parts of the orders modulo the prime powers all agree.
Why the other options are incorrect.
(A) $1/8 = 2^{-3}$ confuses the failure bound with the success bound and uses the wrong exponent.
(B) 1/2 is the bound for $k = 2$ (an RSA modulus), not for three prime factors.
(D) $7/8 = 1 - 2^{-3}$ uses the exponent k instead of $k - 1$.
Key takeaway. For an RSA modulus at least half of the bases work, so a few repetitions suffice.

Q.15 MATCHING LISTS · φ, order, λ, inverse

Match each quantity for $N = 15$ (List-I) with its value (List-II).

List-I		List-II	
(P)	$\varphi(15)$	**(1)**	13
(Q)	order of 4 modulo 15	**(2)**	8
(R)	Carmichael value $\lambda(15) = \mathrm{lcm}(\lambda(3), \lambda(5))$	**(3)**	2
(S)	$7^{-1} \bmod 15$	**(4)**	4

(A) P→4, Q→3, R→2, S→1 **(B)** P→2, Q→3, R→1, S→4
(C) ✓ P→2, Q→3, R→4, S→1 **(D)** P→2, Q→4, R→3, S→1

Answer: (C) Detailed solution

Correct answer. (C) P→2, Q→3, R→4, S→1
Concepts required. φ, λ, orders and inverses are basic invariants of $\mathbb{Z}_N^*$; for $N = 15$ they can be computed by hand.
Relevant formulae. $\varphi(15) = 2 \cdot 4 = 8$; $4^2 = 16 \equiv 1$ so $\mathrm{ord}(4) = 2$; $\lambda(15) = \mathrm{lcm}(2, 4) = 4$; $7 \cdot 13 = 91 \equiv 1$.
Step-by-step explanation.

- $\varphi(15) = 2 \cdot 4 = 8$ (2).
- $4^2 = 16 \equiv 1$: order 2 (3).
- $\lambda(15) = \mathrm{lcm}(2, 4) = 4$ (4): every order divides it.
- $7 \cdot 13 = 91 = 6 \cdot 15 + 1$: $7^{-1} = 13$ (1).

Hence P→2, Q→3, R→4, S→1.
Why the other options are incorrect.

(A) This assigns 4 to $\varphi(15)$; $\varphi(15) = 8$ counts $1, 2, 4, 7, 8, 11, 13, 14$.

(B) This gives $\lambda(15) = 13$ and $7^{-1} = 4$; but $7 \cdot 4 = 28 \equiv 13 \not\equiv 1$.

(D) This swaps the order of 4 and $\lambda(15)$: $4^2 \equiv 1$, so the order is 2, while $\lambda(15) = 4$.

Key takeaway. Orders divide $\lambda(N)$, which divides $\varphi(N)$: here $2 \mid 4 \mid 8$.

Q.16 MATCHING LISTS · Classical factoring complexity

Match each classical algorithm (List-I) with its running time for an n-bit input N (List-II).

List-I		List-II	
(P)	trial division	**(1)**	$\exp\left((64/9)^{1/3}(\ln N)^{1/3}(\ln \ln N)^{2/3}(1+o(1))\right)$ (sub-exponential)
(Q)	Pollard's rho	**(2)**	$O(N^{1/4})$ expected, i.e. $O(2^{n/4})$ (heuristic)
(R)	general number field sieve	**(3)**	$O(n)$ division steps (polynomial)
(S)	Euclidean $\gcd(a, N)$	**(4)**	$O(\sqrt{N}) = O(2^{n/2})$ divisions

(A) P→2, Q→4, R→1, S→3 **(B)** P→4, Q→1, R→2, S→3

(C) P→4, Q→2, R→3, S→1 **(D)** ✓ P→4, Q→2, R→1, S→3

Answer: (D) **Detailed solution**

Correct answer. (D) P→4, Q→2, R→1, S→3

Concepts required. Classical factoring methods range from exponential (trial division, rho) to sub-exponential (GNFS); gcd is polynomial.

Relevant formulae. Trial division $O(2^{n/2})$; Pollard rho $O(2^{n/4})$; GNFS $L_N[1/3, (64/9)^{1/3}]$; Euclid $O(n)$ steps.

Step-by-step explanation.

- trial division tests divisors up to $\sqrt{N}$ (4);
- Pollard's rho finds a factor p in about $\sqrt{p} \le N^{1/4}$ steps (2);
- the GNFS is the fastest known general method, sub-exponential but super-polynomial (1);
- the gcd is polynomial (3) — not a factoring algorithm on its own.

Hence P→4, Q→2, R→1, S→3. Not all classical factoring is exponential.

Why the other options are incorrect.

(A) This swaps trial division and Pollard's rho; rho finds a factor p in about $\sqrt{p} \le N^{1/4}$ steps, faster than trial division.

(B) This gives Pollard's rho the GNFS running time and vice versa; GNFS is the fastest known general method.

(C) This gives the gcd the GNFS time; the Euclidean algorithm is polynomial, $O(n)$ division steps.

Key takeaway. The best classical method is sub-exponential; Shor's algorithm is polynomial — on a fault-tolerant machine.

Q.17 Assertion–Reason · Non-trivial square root ⇒ factor

Assertion (A): If $x^2 \equiv 1 \pmod{N}$ but $x \not\equiv \pm 1 \pmod{N}$, then $\gcd(x-1, N)$ is a non-trivial factor of N.

Reason (R): In that case N divides $(x-1)(x+1)$ but divides neither $x-1$ nor $x+1$.

(A) ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

Concepts required. If $N \mid (x-1)(x+1)$ but N divides neither factor, each prime factor of N is split between them.

Relevant formulae. $x^2 \equiv 1 \Rightarrow N \mid (x-1)(x+1)$; $x \not\equiv \pm 1 \Rightarrow 1 < \gcd(x \pm 1, N) < N$.

Step-by-step explanation. **R is true**: $x^2 - 1 = (x-1)(x+1) \equiv 0$, while $x \not\equiv \pm 1$ means $N \nmid x \mp 1$. **A follows from R**: if $\gcd(x-1, N)$ were 1, then $N \mid (x-1)(x+1)$ would force $N \mid x+1$; if it were N, then $N \mid x-1$ — both excluded. So it is a proper divisor. This is exactly Shor's factor-extraction step with $x = a^{r/2}$. Answer: (A).

Why the other options are incorrect.

(B) is wrong because R is exactly the argument that proves A: a divisor of a product that divides neither factor must share a proper factor with each.

(C) is wrong because R is true: $x^2 - 1 = (x-1)(x+1) \equiv 0$, and $x \not\equiv \pm 1$ means N divides neither factor.

(D) is wrong because A is true: e.g. $x = 4$, $N = 15$ gives $\gcd(3, 15) = 3$.

Key takeaway. This is the heart of the reduction: $x = a^{r/2}$ is a square root of 1.

Q.18 Statement I / II · Euler's theorem vs order

Statement I: If $\gcd(a, N) = 1$, then $a^{\varphi(N)} \equiv 1 \pmod{N}$.

Statement II: The multiplicative order of every a coprime to N equals $\varphi(N)$.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) ✓ Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (C) Detailed solution

Correct answer. **(C)** Statement I is true, but Statement II is false

Concepts required. Euler's theorem gives an exponent that always works; the order is the smallest one and is usually a proper divisor of $\varphi(N)$.

Relevant formulae. $a^{\varphi(N)} \equiv 1$; $\mathrm{ord}(a) \mid \lambda(N) \mid \varphi(N)$; e.g. $\mathrm{ord}_{15}(4) = 2 \neq 8$.

Step-by-step explanation. **I is true** (Euler's theorem). **II is false**: the order only *divides* $\varphi(N)$; e.g. 4 has order 2 modulo 15 while $\varphi(15) = 8$. Finding the actual order is the hard problem that Shor's quantum period finding solves. Answer: (C).

Why the other options are incorrect.

(A) is wrong because Statement II is false: $\mathrm{ord}_{15}(4) = 2$ while $\varphi(15) = 8$.

(B) is wrong because Statement I is Euler's theorem, which holds for every a coprime to N.

(D) is wrong because Statement I is true and Statement II false, not the other way round.

Key takeaway. If orders were always $\varphi(N)$, knowing them would directly reveal $\varphi(N)$; in reality they only divide it.

Paragraph for Questions 19 and 20

Shor's reduction is applied by hand to $N = 91 = 7 \times 13$ with the base $a = 3$ (note $\gcd(3, 91) = 1$). Recall that the order r of a modulo N is the least $r > 0$ with $a^r \equiv 1 \pmod{N}$, and that if r is even and $a^{r/2} \not\equiv -1 \pmod{N}$, then $\gcd(a^{r/2} \pm 1, N)$ are non-trivial factors.

Q.19 PARAGRAPH BASED · Order of 3 mod 91

The order r of 3 modulo 91 is

(A) 72 **(B)** ✓ 6 **(C)** 12 **(D)** 3

Answer: (B) Detailed solution

Correct answer. (B) 6
Concepts required. The order is the first exponent where the powers return to 1.
Relevant formulae. $3, 9, 27, 81, 243 \equiv 61, 183 \equiv 1 \pmod{91}$ ($729 = 8 \cdot 91 + 1$): $r = 6$.
Step-by-step explanation. By CRT, $r = \text{lcm}(r_7, r_{13})$. Mod 7: $3, 2, 6, 4, 5, 1$: order 6. Mod 13: $3, 9, 27 \equiv 1$: order 3. So $r = \text{lcm}(6, 3) = 6$. *Check*: $3^6 = 729 = 8 \cdot 91 + 1$. ($72 = \varphi(91)$ is only a multiple of the order.)
Why the other options are incorrect.
(A) $72 = \varphi(91)$ is an exponent that works, not the least one.
(C) 12 is a multiple of the order ($3^{12} \equiv 1$) but not the least one.
(D) $3^3 = 27 \neq 1$; 3 is half of the order.
Key takeaway. Order = smallest exponent; multiples and $\varphi(N)$ also satisfy $a^x \equiv 1$.

Q.20 PARAGRAPH BASED · Factors of 91 by gcd

The values of $\gcd(3^{r/2} - 1, 91)$ and $\gcd(3^{r/2} + 1, 91)$ are, respectively,

(A) 26 and 28 **(B)** 1 and 91 **(C)** ✓ 13 and 7 **(D)** 7 and 13

Answer: (C) Detailed solution

Correct answer. (C) 13 and 7
Concepts required. With r even and $a^{r/2} \not\equiv -1$, the gcds with $a^{r/2} \mp 1$ give the factors.
Relevant formulae. $3^{r/2} = 27$; $\gcd(26, 91) = 13$, $\gcd(28, 91) = 7$; $7 \cdot 13 = 91$.
Step-by-step explanation. $r = 6$ is even and $3^3 = 27 \not\equiv -1 \equiv 90$. $\gcd(26, 91) = 13$ (since $91 = 7 \cdot 13$, $26 = 2 \cdot 13$) and $\gcd(28, 91) = 7$ ($28 = 4 \cdot 7$). Both factors are found: $91 = 7 \times 13$. The order of the two numbers matters: $a^{r/2} - 1 = 26$ shares 13 with 91.
Why the other options are incorrect.
(A) 26 and 28 are $a^{r/2} \mp 1$ themselves; the gcd with 91 is still needed.
(B) 1 and 91 would be the trivial result for $a^{r/2} \equiv -1$; but $27 \not\equiv 90$.
(D) 7 and 13 are the correct factors in the wrong order: $\gcd(26, 91) = 13$ and $\gcd(28, 91) = 7$.
Key takeaway. The quantum part ends with r; the factors come from two classical gcds.

SECTION B · Quantum Computing Foundations Q.21–40

Q.21 Single-correct MCQ · Uniform superposition

After $H^{\otimes 8}$ acts on $|0\rangle^{\otimes 8}$ (the counting register of the notebook for $N = 15$), the amplitude of each basis state $|x\rangle$ is

(A) $1/256$ **(B)** ✓ $1/16$ **(C)** $1/8$ **(D)** $1/\sqrt{8}$

Answer: (B) Detailed solution

Correct answer. (B) $1/16$

Concepts required. The Hadamard transform of $|0\rangle^{\otimes t}$ is the uniform superposition of all 2^t basis states.

Relevant formulae. $H^{\otimes t}|0\rangle^{\otimes t} = 2^{-t/2}\sum_x |x\rangle$; $t = 8$: amplitude $1/16$, probability $1/256$.

Step-by-step explanation. $H^{\otimes t}|0\rangle^{\otimes t} = \frac{1}{\sqrt{2^t}}\sum_{x=0}^{2^t-1}|x\rangle$. With $t = 8$: $\frac{1}{\sqrt{256}} = \frac{1}{16}$. The probabilities are $\frac{1}{256}$ each. This superposition is the input on which the modular exponentiation acts "for all x at once".

Why the other options are incorrect.

(A) $1/256$ is the probability $|1/16|^2$, not the amplitude.

(C) $1/8$ would be the amplitude for $t = 6$ (2^{-3}); for $t = 8$ it is 2^{-4}.

(D) $1/\sqrt{8}$ is the amplitude for $t = 3$ qubits.

Key takeaway. Amplitudes are $2^{-t/2}$; probabilities 2^{-t}.

Q.22 Single-correct MCQ · Measuring the work register

The state $\frac{1}{\sqrt{8}}\sum_{x=0}^{7}|x\rangle|2^x \bmod 15\rangle$ is prepared and the *second* register is measured with result 4. The probability of this result and the post-measurement state of the first register are

(A) $1/2$; $\frac{1}{\sqrt{2}}(|2\rangle + |6\rangle)$ **(B)** $1/4$; $\frac{1}{\sqrt{2}}(|2\rangle - |6\rangle)$

(C) $1/8$; $|2\rangle$ **(D)** ✓ $1/4$; $\frac{1}{\sqrt{2}}(|2\rangle + |6\rangle)$

Answer: (D) Detailed solution

Correct answer. (D) $1/4$; $\frac{1}{\sqrt{2}}(|2\rangle + |6\rangle)$

Concepts required. Measuring one register of an entangled state keeps only the compatible terms of the other, renormalised.

Relevant formulae. $2^x \bmod 15$: $1, 2, 4, 8, 1, 2, 4, 8$; value 4 at $x = 2, 6$: $P = 2/8$, state $(|2\rangle+|6\rangle)/\sqrt{2}$.

Step-by-step explanation. $2^x \bmod 15$ for $x = 0, \ldots, 7$: $1, 2, 4, 8, 1, 2, 4, 8$. The value 4 occurs for $x = 2$ and $x = 6$, so $P = 2 \times \frac{1}{8} = \frac{1}{4}$, and the first register collapses to $\frac{1}{\sqrt{2}}(|2\rangle + |6\rangle)$ (equal positive amplitudes, renormalised). It is periodic with period $r = 4$ and an unknown offset — the QFT removes the offset from the probabilities.

Why the other options are incorrect.

(A) $1/2$ counts two of the four values as matching; only $x = 2, 6$ of 8 terms give 4.

(B) The amplitudes before the measurement are all $+1/\sqrt{8}$; no minus sign can appear.

(C) $1/8$ and $|2\rangle$ forget that the function is periodic: $x = 6$ gives $2^6 = 64 \equiv 4$ as well.

Key takeaway. The collapsed first register is periodic with period r and a random offset.

Q.23 Single-correct MCQ · Unitarity of U_a

Why does the operator $U_a|y\rangle = |ay \bmod N\rangle$ (for $0 \le y < N$; identity for $y \ge N$) require $\gcd(a, N) = 1$ to be a valid quantum gate?

(A) because the eigenvalues of U_a must be real

(B) because a quantum gate must map $|0\rangle$ to $|0\rangle$

(C) ✓ only then is $y \mapsto ay \bmod N$ a permutation of $\{0, \ldots, N-1\}$, so U_a is a permutation matrix and hence unitary

(D) because otherwise U_a would not be Hermitian

Answer: (C) **Detailed solution**

Correct answer. **(C)** only then is $y \mapsto ay \bmod N$ a permutation of $\{0, \ldots, N-1\}$, so U_a is a permutation matrix and hence unitary

Concepts required. A permutation matrix is unitary; a non-injective map of basis states is not.

Relevant formulae. $\gcd(a, N) = 1 \iff a$ invertible mod $N \iff y \mapsto ay \bmod N$ bijective on $\{0, \ldots, N-1\}$.

Step-by-step explanation. If $\gcd(a, N) = d > 1$, two inputs collide: e.g. $a = 3$, $N = 15$: $3 \cdot 0 \equiv 3 \cdot 5 \equiv 0$. A non-injective map cannot be unitary. For $\gcd(a, N) = 1$, a is invertible modulo N, the map is a bijection, and a permutation matrix satisfies $U^\dagger U = I$. Gates need not be Hermitian, and their eigenvalues are complex phases.

Why the other options are incorrect.

(A) Unitary eigenvalues are phases $e^{i\phi}$; U_a has complex eigenvalues $e^{2\pi i s/r}$.

(B) Gates need not fix $|0\rangle$; U_a happens to fix $|0\rangle$, but that is not the requirement.

(D) Quantum gates must be unitary, not Hermitian; U_a is generally not Hermitian (it cycles states).

Key takeaway. Coprimality makes modular multiplication reversible, hence implementable as a quantum gate.

Q.24 Single-correct MCQ · Eigenvalues of U_a

On the subspace spanned by $\{|1\rangle, |2\rangle, |4\rangle, |8\rangle\}$, the operator $U_2|y\rangle = |2y \bmod 15\rangle$ has the eigenvalues

(A) $1, 2, 4, 8$

(B) $1, -1$ (each twice)

(C) $e^{2\pi i/15}, e^{4\pi i/15}, e^{8\pi i/15}, e^{16\pi i/15}$

(D) ✓ $1, i, -1, -i$

Answer: (D) **Detailed solution**

Correct answer. **(D)** $1, i, -1, -i$

Concepts required. A cyclic permutation of length r has the r-th roots of unity as eigenvalues.

Relevant formulae. U_2: $1 \to 2 \to 4 \to 8 \to 1$, $r = 4$; eigenvalues $e^{2\pi i s/4} = 1, i, -1, -i$.

Step-by-step explanation. U_2 cycles $|1\rangle \to |2\rangle \to |4\rangle \to |8\rangle \to |1\rangle$: a cyclic shift of length $r = 4$, so $U_2^4 = I$ there and the eigenvalues are the fourth roots of unity $e^{2\pi i s/4}$, $s = 0, 1, 2, 3$. In general the eigenvalues on the orbit of $|1\rangle$ are $e^{2\pi i s/r}$: the order r is encoded in the eigenphases s/r — this is what phase estimation reads out.

Why the other options are incorrect.

(A) $1, 2, 4, 8$ are the basis labels on the orbit, not eigenvalues; unitary eigenvalues have modulus 1.

(B) ± 1 would require $U_2^2 = I$, but $U_2^2|1\rangle = |4\rangle \neq |1\rangle$.

(C) Phases with denominator 15 use N instead of the order $r = 4$.

Key takeaway. The order is hidden in the eigenphases s/r of U_a.

Q.25 Single-correct MCQ · $|1\rangle$ as sum of eigenvectors

With $|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} |a^k \bmod N\rangle$, the superposition $\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle$ equals

(A) $|a\rangle$ **(B)** ✓ $|1\rangle$

(C) $|0\rangle$ **(D)** $\frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} |y\rangle$

Answer: (B) Detailed solution

Correct answer. (B) $|1\rangle$

Concepts required. The eigenvectors $|u_s\rangle$ are Fourier transforms of the orbit; summing over s inverts the transform.

Relevant formulae. $\sum_{s=0}^{r-1} e^{-2\pi i s k/r} = r\,\delta_{k,0}$, so $\frac{1}{\sqrt{r}} \sum_s |u_s\rangle = |a^0\rangle = |1\rangle$.

Step-by-step explanation. $\frac{1}{\sqrt{r}} \sum_s |u_s\rangle = \frac{1}{r} \sum_k \left(\sum_s e^{-2\pi i s k/r} \right) |a^k\rangle = \sum_k \delta_{k,0} |a^k\rangle = |a^0\rangle = |1\rangle$, because the sum of r-th roots of unity vanishes unless $k \equiv 0$. That is why the work register is initialised to $|1\rangle$: it is an equal superposition of all eigenvectors, and QPE returns a random s/r.

Why the other options are incorrect.

(A) $|a\rangle = |a^1\rangle$ would need the phase factors $e^{2\pi i s/r}$ inserted before summing.

(C) $|0\rangle$ is outside the orbit $\{a^k \bmod N\}$ of $|1\rangle$, so it cannot be a combination of the $|u_s\rangle$.

(D) The uniform superposition over all residues is not in the span of one orbit (it includes non-coprime values).

Key takeaway. Initialising the work register to $|1\rangle$ prepares all eigenvectors at once, so QPE yields a random s/r.

Q.26 Single-correct MCQ · Phase kickback

A control qubit in $|+\rangle$ controls a unitary U acting on its eigenvector $|u\rangle$, $U|u\rangle = e^{i\phi}|u\rangle$, with $\phi = 2\pi/3$. Afterwards, the expectation value $\langle X \rangle$ of the control qubit is

(A) $1/2$ **(B)** 1 **(C)** $-\sqrt{3}/2$ **(D)** ✓ $-1/2$

Answer: (D) Detailed solution

Correct answer. (D) $-1/2$

Concepts required. Phase kickback moves the eigenphase of the target onto the $|1\rangle$ branch of the control.

Relevant formulae. $|+\rangle|u\rangle \mapsto \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)|u\rangle$; $\langle X \rangle = \cos\phi$, $\langle Y \rangle = \sin\phi$.

Step-by-step explanation. Phase kickback: $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|u\rangle \mapsto \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)|u\rangle$ — the work register is unchanged, the phase moves to the control. For $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)$, $\langle X \rangle = \cos\phi = \cos\frac{2\pi}{3} = -\frac{1}{2}$ (and $\langle Y \rangle = \sin\phi = \frac{\sqrt{3}}{2}$).

Why the other options are incorrect.

(A) $1/2 = \cos(\pi/3)$ uses half of the phase.

(B) $\langle X \rangle = 1$ is the value before the controlled gate; the kickback rotates the state.

(C) $-\sqrt{3}/2$ mixes up sine and cosine; $\sin(2\pi/3) = +\sqrt{3}/2$ is $\langle Y \rangle$.

Key takeaway. The control qubit records the phase; the eigenvector target is untouched.

Q.27 Single-correct MCQ · Phases of U^{2^j}

In phase estimation, counting qubit j controls U^{2^j}. If $U|u\rangle = e^{2\pi i/6}|u\rangle$, the phase kicked back onto counting qubit $j = 2$ is

(A) 2π **(B)** $\pi/3$ **(C)** ✓ $4\pi/3$ **(D)** $2\pi/3$

Answer: (C) Detailed solution

Correct answer. (C) $4\pi/3$
Concepts required. Applying U^{2^j} to an eigenvector multiplies its phase by 2^j.
Relevant formulae. Phase on qubit j: $2\pi\varphi \cdot 2^j$; $\varphi = 1/6$, $j = 2$: $2\pi \cdot 4/6 = 4\pi/3$.
Step-by-step explanation. $U^{2^j}|u\rangle = e^{2\pi i \cdot 2^j/6}|u\rangle$; for $j = 2$: $2\pi \cdot \frac{4}{6} = \frac{4\pi}{3}$ (equivalently $-\frac{2\pi}{3}$). Each counting qubit receives the phase doubled relative to the previous one — the binary-fraction pattern that the inverse QFT decodes.
Why the other options are incorrect.
(A) 2π would correspond to U^6; $2^j = 4$, not 6.
(B) $\pi/3 = 2\pi/6$ is the phase for $j = 0$ (U itself).
(D) $2\pi/3$ is the phase for $j = 1$ (U^2).
Key takeaway. Counting qubit j carries the binary digit pattern $2^j\varphi$ mod 1.

Q.28 Single-correct MCQ · Hadamard transform signs

$(H \otimes H)|11\rangle$ equals

(A) ✓ $\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)$ **(B)** $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$
(C) $\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle)$ **(D)** $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

Answer: (A) Detailed solution

Correct answer. (A) $\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)$
Concepts required. $H^{\otimes n}|x\rangle = 2^{-n/2}\sum_y (-1)^{x\cdot y}|y\rangle$.
Relevant formulae. $x = 11$: signs $(-1)^{y_1+y_2}$: $+, -, -, +$ for $00, 01, 10, 11$.
Step-by-step explanation. $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, so $H|1\rangle \otimes H|1\rangle = \frac{1}{2}(|0\rangle - |1\rangle)(|0\rangle - |1\rangle) = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)$. In general $H^{\otimes n}|x\rangle = 2^{-n/2}\sum_y (-1)^{x\cdot y}|y\rangle$: signs depend on the parity of $x \cdot y$.
Why the other options are incorrect.
(B) A Bell-type state cannot arise from local Hadamards on a product state.
(C) The last sign is $(-1)^{1+1} = +1$, not -1.
(D) All plus signs is $(H \otimes H)|00\rangle$.
Key takeaway. Hadamard transforms encode parities in signs — the simplest Fourier transform.

Q.29 Single-correct MCQ · Two-path interference

Two computational paths lead to the same outcome with amplitudes $\frac{1}{2}$ and $\frac{1}{2}e^{i\theta}$. For $\theta = 2\pi/3$ the probability of that outcome is

(A) 0 **(B)** 1/2 **(C)** ✓ 1/4 **(D)** 3/4

Answer: (C) Detailed solution

Correct answer. **(C)** 1/4

Concepts required. Probabilities come from the squared modulus of the *sum* of amplitudes of indistinguishable paths.

Relevant formulae. $\left|\frac{1}{2}(1+e^{i\theta})\right|^2 = \cos^2(\theta/2)$; $\theta = 2\pi/3$: $\cos^2(\pi/3) = 1/4$.

Step-by-step explanation. Amplitudes add before squaring: $\left|\frac{1}{2}(1+e^{i\theta})\right|^2 = \frac{1}{4}|1+e^{i\theta}|^2 = \frac{1}{4}(2+2\cos\theta) = \cos^2\frac{\theta}{2} = \cos^2\frac{\pi}{3} = \frac{1}{4}$. Interference concentrates probability where the phases agree; in Shor's algorithm the inverse QFT arranges this for $y \approx k2^t/r$.

Why the other options are incorrect.

(A) 0 needs $\theta = \pi$ (complete destructive interference).

(B) 1/2 adds probabilities ($\frac{1}{4} + \frac{1}{4}$) instead of amplitudes, ignoring interference.

(D) $3/4 = \sin^2(\pi/3)$ is the probability of the other output port.

Key takeaway. Interference, not parallelism alone, makes the right answer likely.

Q.30 Single-correct MCQ · Reversible modular exponentiation

Why does Shor's circuit compute $|x\rangle|1\rangle \mapsto |x\rangle|a^x \bmod N\rangle$ instead of $|x\rangle \mapsto |a^x \bmod N\rangle$ in place?

(A) ✓ the map $x \mapsto a^x \bmod N$ is not injective (x and $x+r$ give the same value), so it cannot be a unitary on a single register; keeping x makes it reversible

(B) because a single register cannot hold numbers larger than N

(C) because the Hadamard gates must act on a different register than the QFT

(D) because measurement requires two registers

Answer: (A) Detailed solution

Correct answer. **(A)** the map $x \mapsto a^x \bmod N$ is not injective (x and $x+r$ give the same value), so it cannot be a unitary on a single register; keeping x makes it reversible

Concepts required. Unitary operations are invertible; a many-to-one function must keep its input to be reversible.

Relevant formulae. $a^{x+r} \equiv a^x$; $|x\rangle|y\rangle \mapsto |x\rangle|y\,a^x \bmod N\rangle$ is reversible.

Step-by-step explanation. Quantum gates are unitary, hence invertible. Since $a^{x+r} \equiv a^x$, an in-place map would send different inputs to the same output. Keeping the input x and writing the result into a second register (here by multiplying the work register $|1\rangle$ by a^x) gives a reversible map and entangles the registers: $\sum_x |x\rangle|a^x \bmod N\rangle$.

Why the other options are incorrect.

(B) Register size is not the issue: $a^x \bmod N < N$ always fits in n qubits.

(C) The Hadamards and the QFT both act on the counting register; this is not why two registers are used.

(D) Measurement can be performed on any register; it does not require a second one.

Key takeaway. Reversibility forces the input to be kept, which entangles the two registers.

Q.31 Single-correct MCQ · CNOT cost of a Fredkin gate

A controlled-SWAP (Fredkin) gate is compiled as $\mathrm{CNOT}_{b\to a} \cdot \mathrm{Toffoli}_{c,a\to b} \cdot \mathrm{CNOT}_{b\to a}$, and each Toffoli uses the standard decomposition with 6 CNOTs. The number of CNOTs per controlled-SWAP is

(A) 9 **(B)** ✓ 8 **(C)** 3 **(D)** 6

Answer: (B) Detailed solution

Correct answer. **(B)** 8
Concepts required. A Fredkin gate can be built from two CNOTs around a Toffoli; a Toffoli needs 6 CNOTs in the standard decomposition.
Relevant formulae. $\mathrm{CSWAP}_{c;a,b} = \mathrm{CX}_{b\to a}\,\mathrm{CCX}_{c,a\to b}\,\mathrm{CX}_{b\to a}$: $1+6+1=8$.
Step-by-step explanation. $1+6+1=8$ CNOTs (plus single-qubit H, T, $T^\dagger$ gates). A plain SWAP needs 3 CNOTs, but the control turns the middle CNOT into a Toffoli. Gate-level modular multipliers are built from many such gates, which is why compiled Shor circuits are much larger than the notebook's matrix gates.
Why the other options are incorrect.

(A) $9 = 3 \times 3$ treats every CNOT of a SWAP as a 3-CNOT gate; only the middle CNOT must become a (6-CNOT) Toffoli.

(C) 3 CNOTs make an uncontrolled SWAP; the control requires the Toffoli.

(D) 6 counts only the Toffoli and forgets the two outer CNOTs.

Key takeaway. Controlled arithmetic is costly: every controlled SWAP is a Toffoli plus two CNOTs.

Q.32 Single-correct MCQ · Implicit measurement

`Shor.ipynb` measures only the counting register. Compared with also measuring the work register first, the distribution of the counting outcome y is

(A) different: the peaks become twice as wide

(B) undefined, because entangled registers cannot be measured separately

(C) ✓ exactly the same: a measurement of the other register that is never used cannot change the marginal statistics of y

(D) different: without measuring the work register no periodic structure appears

Answer: (C) Detailed solution

Correct answer. **(C)** exactly the same: a measurement of the other register that is never used cannot change the marginal statistics of y
Concepts required. Measuring a subsystem that is never used again does not change the reduced state of the rest.
Relevant formulae. $\rho_{\text{count}} = \mathrm{Tr}_{\text{work}}\,|\Psi\rangle\langle\Psi| = \sum_w P(w)\rho_{\text{count}|w}$.
Step-by-step explanation. The work register is not acted on after the controlled multiplications. Measuring it (or not) only changes which conditional state the counting register is in; averaged over the unseen result, the reduced state of the counting register — and hence $P(y)$ — is identical (principle of implicit measurement). Textbook derivations "measure the second register" only to simplify the algebra.
Why the other options are incorrect.

(A) Peak widths depend on t and r, not on whether the idle work register is measured.

(B) Entangled registers can be measured separately; the marginal statistics are well defined.

(D) The periodic structure is in the entangled state already; the measurement of the work register only simplifies textbook algebra.

Key takeaway. Principle of implicit (deferred) measurement: unused registers may be measured or ignored.

Q.33 Single-correct MCQ · Notebook qubit count

In `Shor.ipynb` the work register has $n =$ `N.bit_length()` qubits and the counting register $t = 2n$. The total number of qubits for $N = 143$ is

(A) ✓ 24 **(B)** 21 **(C)** 22 **(D)** 16

Answer: (A) Detailed solution

Correct answer. **(A)** 24
Concepts required. The notebook uses $n = \lceil \log_2(N+1) \rceil$ work qubits and $t = 2n$ counting qubits: $3n$ in total.
Relevant formulae. $143 = 10001111_2$: $n = 8$, $3n = 24$.
Step-by-step explanation. $143 = 10001111_2$ has 8 bits, so $n = 8$, $t = 16$ and $t + n = 24$ (the notebook's comment lists "143 -> 24"). 16 forgets the work register; $22 = 2\lceil \log_2 143 \rceil + 6$ has no basis.
Why the other options are incorrect.
(B) 21 would be $3 \cdot 7$, but $143 > 127 = 2^7 - 1$ needs 8 bits.
(C) 22 does not follow from $t = 2n$ with $n = 8$; no formula of the notebook gives it.
(D) 16 is only the counting register $t = 2n$.
Key takeaway. Qubits scale as $3n$ here: three per bit of N.

Q.34 Single-correct MCQ · State-vector memory

The memory of a double-precision state vector (16 bytes per amplitude) for the 24-qubit circuit of the previous question is about

(A) 16.8 MB **(B)** 384 bytes **(C)** 4.3 GB **(D)** ✓ 268 MB

Answer: (D) Detailed solution

Correct answer. **(D)** 268 MB
Concepts required. A state vector of q qubits has 2^q complex amplitudes of 16 bytes (complex128).
Relevant formulae. $M = 16 \cdot 2^q$ bytes; $q = 24$: 268 435 456 B $\approx$ 268 MB.
Step-by-step explanation. $16 \times 2^{24} = 268\,435\,456$ bytes $\approx$ 268 MB (the notebook's table). $384 = 16 \times 24$ is linear; 16.8 MB is $2^{20} \times 16$ (20 qubits); 4.3 GB corresponds to 28 qubits. Each extra bit of N adds 3 qubits, i.e. multiplies the memory by 8.
Why the other options are incorrect.
(A) 16.8 MB is $16 \cdot 2^{20}$, the memory of 20 qubits.
(B) 384 bytes grows linearly with the number of qubits; state vectors grow exponentially.
(C) 4.3 GB is $16 \cdot 2^{28}$, the memory of 28 qubits.
Key takeaway. Each extra qubit doubles the memory; each extra bit of N multiplies it by 8.

Q.35 MATCHING LISTS · Roles of circuit blocks

Match each operation of the order-finding circuit (List-I) with its role (List-II).

List-I		List-II	
(P)	H on every counting qubit	**(1)**	turns the phases $2\pi \cdot 2^j s/r$ into a binary estimate $y \approx 2^t s/r$
(Q)	controlled $U_{a^{2^j} \bmod N}$	**(2)**	prepares the work register in $\lvert 1\rangle$, an equal superposition of the eigenvectors $\lvert u_s\rangle$
(R)	inverse QFT on the counting register	**(3)**	creates the uniform superposition over all exponents x
(S)	X on the last work qubit	**(4)**	performs modular exponentiation / kicks the eigenphases back onto the counting qubits

(A) P→3, Q→1, R→4, S→2
(B) ✓ P→3, Q→4, R→1, S→2
(C) P→3, Q→4, R→2, S→1
(D) P→4, Q→3, R→1, S→2

Answer: (B) Detailed solution

Correct answer. **(B)** P→3, Q→4, R→1, S→2
Concepts required. Order finding: superposition, controlled modular exponentiation (kickback), inverse QFT, measurement.
Relevant formulae. $\frac{1}{\sqrt{2^t}} \sum_x |x\rangle |1\rangle \to \frac{1}{\sqrt{2^t}} \sum_x |x\rangle |a^x\rangle \to$ peaks at $y \approx k2^t/r$.
Step-by-step explanation.

- Hadamards: superposition over x (3);
- controlled powers of U_a: $|x\rangle |1\rangle \mapsto |x\rangle |a^x\rangle$, equivalently phase kickback (4);
- inverse QFT: interference that converts the periodic phase pattern into peaks at $y \approx k2^t/r$ (1);
- X on the least significant work qubit: $|0001\rangle = |1\rangle$ (2).

Hence P→3, Q→4, R→1, S→2.
Why the other options are incorrect.
(A) This gives the controlled multiplications the decoding role of the inverse QFT and the QFT the exponentiation.
(C) This swaps the roles of the inverse QFT and the X gate; X only prepares $|1\rangle$ in the work register.
(D) This gives the Hadamards the role of modular exponentiation and vice versa; the Hadamards only create the superposition.
Key takeaway. Know what each block does: it is the basis for every resource and error discussion.

Q.36 MATCHING LISTS · Phase gates $R_1(\lambda)$

The phase gate is $R_1(\lambda) = \mathrm{diag}(1, e^{i\lambda})$ (controlled versions appear in the QFT). Match each angle (List-I) with the named gate (List-II).

List-I		List-II	
(P)	$R_1(\pi)$	**(1)**	$S^\dagger$
(Q)	$R_1(\pi/2)$	**(2)**	T
(R)	$R_1(\pi/4)$	**(3)**	Z
(S)	$R_1(-\pi/2)$	**(4)**	S

(A) P→3, Q→4, R→1, S→2
(B) ✓ P→3, Q→4, R→2, S→1
(C) P→3, Q→2, R→4, S→1
(D) P→4, Q→3, R→2, S→1

Answer: (B) Detailed solution

Correct answer. (B) P→3, Q→4, R→2, S→1
Concepts required. $R_1(\lambda) = \mathrm{diag}(1, e^{i\lambda})$ contains Z, S, T and their inverses as special cases.
Relevant formulae. $e^{i\pi} = -1$, $e^{i\pi/2} = i$, $e^{i\pi/4}$, $e^{-i\pi/2} = -i$.
Step-by-step explanation. $R_1(\pi) = \mathrm{diag}(1,-1) = Z$ (3); $R_1(\pi/2) = \mathrm{diag}(1,i) = S$ (4); $R_1(\pi/4) = \mathrm{diag}(1, e^{i\pi/4}) = T$ (2); $R_1(-\pi/2) = \mathrm{diag}(1,-i) = S^\dagger$ (1). The inverse QFT of the notebook uses `cr1(-pi/2**(k-j))`: controlled $S^\dagger$, $T^\dagger$, ...
Why the other options are incorrect.
(A) This exchanges T and $S^\dagger$; $R_1(-\pi/2)$ has $-i$ on the diagonal, which is $S^\dagger$.
(C) This exchanges S and T: $R_1(\pi/2) = \mathrm{diag}(1,i) = S$, while T has $e^{i\pi/4}$.
(D) This assigns $R_1(\pi)$ to S; $R_1(\pi) = \mathrm{diag}(1,-1) = Z$.
Key takeaway. The inverse QFT uses controlled $R_1(-\pi/2^k)$: controlled $S^\dagger$, $T^\dagger$ and finer rotations.

Q.37 ASSERTION–REASON · Kickback and controls

Assertion (A): When the target register is in an eigenvector of U, a controlled-U gate leaves the target register unchanged.

Reason (R): Controlled gates never change the state of their control qubit.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) ✓ A is true, but R is false
(D) A is false, but R is true

Answer: (C) Detailed solution

Correct answer. (C) A is true, but R is false
Concepts required. Phase kickback: a controlled-U on an eigenvector changes the control (in superposition), not the target.
Relevant formulae. $(\alpha|0\rangle + \beta|1\rangle)|u\rangle \mapsto (\alpha|0\rangle + \beta e^{i\phi}|1\rangle)|u\rangle$.
Step-by-step explanation. **A is true**: $|c\rangle|u\rangle \mapsto |c\rangle\, e^{ic\phi}|u\rangle$ — the target stays $|u\rangle$, only a phase appears. **R is false**: that phase belongs to the control branch $|1\rangle$, so a control in superposition changes, $|+\rangle \mapsto \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)$ (phase kickback). This is precisely how phase estimation and Shor's algorithm extract information. Answer: (C).
Why the other options are incorrect.
(A) is wrong because R is false: the control's relative phase changes, as phase kickback shows.

(B) is wrong for the same reason: R is false, so it cannot be true but irrelevant.

(D) is wrong because A is true: the target remains the eigenvector $|u\rangle$.

Key takeaway. In QPE the information flows from target to control — the opposite of the classical intuition.

Q.38 Statement I / II · Parallelism vs interference

Statement I: Because the circuit evaluates $a^x \bmod N$ for all x in superposition, measuring the registers directly reveals the period r.

Statement II: Interference produced by the inverse QFT is needed to turn the periodic structure into high-probability measurement outcomes.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) ✓ Statement I is false, but Statement II is true

Answer: (D) Detailed solution

Correct answer. (D) Statement I is false, but Statement II is true

Concepts required. Quantum parallelism produces a superposition, but a measurement returns one sample; interference extracts global properties.

Relevant formulae. Measuring $\sum_x |x\rangle\,|f(x)\rangle$ gives one pair $(x, f(x))$; the inverse QFT concentrates amplitude at $y \approx k2^t/r$.

Step-by-step explanation. **I is false**: a measurement returns one random x and one value $a^x \bmod N$ — no more information than a single classical evaluation. Quantum parallelism alone does not solve factoring. **II is true**: the inverse QFT makes the amplitudes interfere constructively near $y = k2^t/r$, so that one measurement carries information about r. Answer: (D).

Why the other options are incorrect.

(A) is wrong because Statement I is false: a direct measurement yields one random evaluation, not the period.

(B) is wrong because Statement II is true: interference via the inverse QFT is the essential step.

(C) has the truth values reversed: I is false and II is true.

Key takeaway. Parallelism alone does not solve factoring; interference does the work.

Paragraph for Questions 39 and 40

Quantum phase estimation (QPE) with $t = 3$ counting qubits is applied to a unitary U and an eigenvector $|u\rangle$ with $U|u\rangle = e^{2\pi i\varphi}|u\rangle$, $0 \le \varphi < 1$. After the Hadamards, the controlled-U^{2^j} gates and the inverse QFT, the counting register is measured and gives $y \in \{0, \dots, 7\}$ with probability

$$P(y) = \left|\frac{1}{8}\sum_{x=0}^{7} e^{2\pi i x(\varphi - y/8)}\right|^2 = \frac{\sin^2\left(8\pi(\varphi - y/8)\right)}{64\,\sin^2\left(\pi(\varphi - y/8)\right)} \quad (P = 1 \text{ if } \varphi = y/8).$$

The estimate of the phase is $y/8$; y is written in binary with the most significant bit first.

Q.39 PARAGRAPH BASED · QPE: exact phase

For $\varphi = 3/8$ the measurement gives

(A) ✓ $y = 3$ (bits 011) with probability 1
(B) $y = 6$ (bits 110) with probability 1
(C) $y = 3$ with probability 1/2
(D) $y = 3$ or $y = 4$ with probability 1/2 each

Answer: (A) Detailed solution

Correct answer. **(A)** $y = 3$ (bits 011) with probability 1
Concepts required. QPE is exact when the phase has a t-bit binary expansion.
Relevant formulae. $\varphi = 3/8 = 0.011_2$; $P(y) = 1$ for $y = 8\varphi = 3$.
Step-by-step explanation. $\varphi = 3/8 = 0.011_2$ is exactly representable with three bits, so all terms in the sum have phase 0 for $y = 3$: $P(3) = 1$, and $P(y) = 0$ for every other y (the geometric sum of eighth roots of unity vanishes). 110 is the bit-reversed string — the inverse QFT's final SWAPs make the most significant bit come first.
Why the other options are incorrect.
(B) 110 is the bit-reversed string; the notebook's inverse QFT includes the SWAPs, so $y = 3$ is read MSB first.
(C) For an exactly representable phase all amplitude interferes constructively at $y = 3$; probability 1, not 1/2.
(D) Splitting between 3 and 4 happens for φ halfway between grid points (7/16), not for 3/8.
Key takeaway. If r divides 2^t, Shor's outcomes are exactly $s2^t/r$ (as for $N = 15$).

Q.40 PARAGRAPH BASED · QPE: inexact phase

For $\varphi = 1/3$ the most likely outcome and its probability are

(A) ✓ $y = 3$ with $P \approx 0.69$
(B) $y = 3$ with $P \approx 0.41$
(C) $y = 2$ or 3 with $P = 1/2$ each
(D) $y = 3$ with $P = 1$

Answer: (A) Detailed solution

Correct answer. **(A)** $y = 3$ with $P \approx 0.69$
Concepts required. For phases that are not multiples of 2^{-t} the outcome distribution is a peak around $2^t\varphi$.
Relevant formulae. $\delta = 1/3 - 3/8 = -1/24$; $P(3) = \sin^2(\pi/3)/(64\sin^2(\pi/24)) \approx 0.688$.
Step-by-step explanation. $8\varphi = 2.67$, nearest integer $y = 3$, $\delta = \varphi - 3/8 = -\frac{1}{24}$. $P(3) = \frac{\sin^2(8\pi/24)}{64\sin^2(\pi/24)} = \frac{\sin^2(\pi/3)}{64\sin^2(7.5°)} = \frac{0.75}{64\times 0.017037} = \frac{0.75}{1.0904} \approx 0.688$. Because 1/3 has no finite binary expansion, the probability spreads over neighbouring y; $\frac{4}{\pi^2} \approx 0.41$ is only the worst-case lower bound for the nearest outcome.
Why the other options are incorrect.
(B) $4/\pi^2 \approx 0.405$ is only the worst-case lower bound for the nearest outcome, attained when φ lies

halfway.

(C) $P(2) \approx 0.17$ and $P(3) \approx 0.69$: $8\varphi = 2.67$ is closer to 3 than to 2, so the two are far from equal.

(D) Probability 1 needs an exactly representable phase; $1/3$ has the infinite binary expansion $0.0101\ldots_2$.

Key takeaway. Peaks have finite width; continued fractions turn the nearest y into s/r.

SILICO FELLER

SECTION C · Shor's Algorithm Fundamentals Q.41–60

Q.41 Single-correct MCQ · Steps of Shor's algorithm

Which sequence describes Shor's factoring algorithm for an odd composite N that is not a prime power?

(A) find the order r by quantum phase estimation; the factors are r and N/r

(B) apply the QFT to $|N\rangle$; measure; the outcome is a factor of N

(C) pick random a; measure $a^x \bmod N$ in superposition; the most frequent value is a factor

(D) ✓ pick random a; if $\gcd(a, N) > 1$ stop (factor found); find the order r of a by quantum phase estimation and continued fractions; if r is even and $a^{r/2} \not\equiv -1$, output $\gcd(a^{r/2} \pm 1, N)$; otherwise repeat

Answer: (D) Detailed solution

Correct answer. (D) pick random a; if $\gcd(a, N) > 1$ stop (factor found); find the order r of a by quantum phase estimation and continued fractions; if r is even and $a^{r/2} \not\equiv -1$, output $\gcd(a^{r/2} \pm 1, N)$; otherwise repeat

Concepts required. Shor's algorithm is a classical reduction from factoring to order finding, with a quantum subroutine for the order only.

Relevant formulae. $r = \mathrm{ord}_N(a)$; if $2 \mid r$ and $a^{r/2} \not\equiv -1$: $p = \gcd(a^{r/2}-1, N)$, $q = \gcd(a^{r/2}+1, N)$.

Step-by-step explanation. The quantum computer is used only for *order finding*; everything else — the gcd test, continued fractions, the checks on r and the final $\gcd(a^{r/2} \pm 1, N)$ — is classical. Option 3 confuses the order with a factor (for $N = 15$, $a = 7$: $r = 4$ does not divide 15).

Why the other options are incorrect.

(A) The order is not a factor: for $N = 15$, $a = 7$, $r = 4 \nmid 15$; the factors come from $\gcd(a^{r/2} \pm 1, N)$.

(B) A QFT of the number N itself produces no factor; Shor applies the QFT to the counting register after modular exponentiation.

(C) Measuring $a^x \bmod N$ gives one random value of the function; no interference step and no factor appear.

Key takeaway. Quantum part: period finding. Classical part: everything else.

Q.42 Single-correct MCQ · $N = 15$, $a = 2$

For $N = 15$ and $a = 2$, the order r and the factors obtained from $\gcd(a^{r/2} \pm 1, N)$ are

(A) $r = 4$; 1 and 15

(B) ✓ $r = 4$; $\gcd(3, 15) = 3$, $\gcd(5, 15) = 5$

(C) $r = 2$; 3 and 5

(D) $r = 8$; 3 and 5

Answer: (B) Detailed solution

Correct answer. (B) $r = 4$; $\gcd(3, 15) = 3$, $\gcd(5, 15) = 5$

Concepts required. For a small N the order can be read off the list of powers; the gcds then give the factors.

Relevant formulae. $2, 4, 8, 16 \equiv 1 \pmod{15}$: $r = 4$; $2^2 = 4$: $\gcd(3, 15) = 3$, $\gcd(5, 15) = 5$.

Step-by-step explanation. $2^1, 2^2, 2^3, 2^4 \equiv 2, 4, 8, 1$, so $r = 4$. $2^{r/2} = 4$, $4 \not\equiv -1 \equiv 14$. $\gcd(4 - 1, 15) = 3$ and $\gcd(4 + 1, 15) = 5$, and $3 \times 5 = 15$. $r = 8$ is a multiple of the order, not the order (the smallest positive exponent).

Why the other options are incorrect.

(A) 1 and 15 are the trivial gcds that occur only for $a^{r/2} \equiv -1$; here $4 \not\equiv 14$.

(C) $2^2 = 4 \neq 1$, so $r \neq 2$; the order is 4 (the factors are right, but for the wrong reason).

(D) 8 is a multiple of the order; the order is the smallest exponent with $a^r \equiv 1$.

Key takeaway. $N = 15$, $a = 2$ behaves like the notebook default $a = 7$: $r = 4$, factors 3×5.

Q.43 Single-correct MCQ · Failure for $a = N - 1$

For $N = 15$ and $a = 14$, the order finding succeeds perfectly. The classical post-processing then

(A) succeeds with factors 3 and 5

(B) fails because r is odd

(C) ✓ fails: $r = 2$ but $14^1 \equiv -1 \pmod{15}$, so $\gcd(14 \pm 1, 15)$ gives only the trivial factors 15 and 1; another a is needed

(D) fails because $\gcd(14, 15) = 1$

Answer: (C) Detailed solution

Correct answer. **(C)** fails: $r = 2$ but $14^1 \equiv -1 \pmod{15}$, so $\gcd(14 \pm 1, 15)$ gives only the trivial factors 15 and 1; another a is needed

Concepts required. The reduction fails when $a^{r/2} \equiv -1 \pmod{N}$, even if the order is found correctly.

Relevant formulae. $14 \equiv -1$, $14^2 \equiv 1$: $r = 2$, $a^{r/2} = 14 \equiv -1$; $\gcd(13, 15) = 1$, $\gcd(15, 15) = 15$.

Step-by-step explanation. $14^2 = 196 = 13 \cdot 15 + 1$, so $r = 2$ (even). But $a^{r/2} = 14 \equiv -1$: $\gcd(13, 15) = 1$ and $\gcd(15, 15) = 15$ — trivial. For any N, $a = N - 1$ is always such a failing base. $\gcd(a, N) = 1$ is required, not a reason for failure.

Why the other options are incorrect.

(A) 3 and 5 cannot appear: with $a^{r/2} \equiv -1$ both gcds are trivial.

(B) $r = 2$ is even; the odd-order failure is a different case (e.g. $a = 4$ modulo 21).

(D) $\gcd(a, N) = 1$ is the normal case required for order finding, not a cause of failure.

Key takeaway. $a = N - 1$ always fails: a failed attempt must be reported, then another base tried.

Q.44 Single-correct MCQ · Odd order: retry

For $N = 21$ and $a = 4$, the outcome of the classical checks is

(A) ✓ $r = 3$ is odd, so $a^{r/2}$ is not an integer power; the attempt fails and a new a is chosen

(B) $r = 3$, factors $\gcd(4^1 - 1, 21) = 3$ and $\gcd(4^1 + 1, 21) = 1$

(C) $\gcd(4, 21) = 7$, a lucky factor

(D) $r = 6$, factors 3 and 7

Answer: (A) Detailed solution

Correct answer. **(A)** $r = 3$ is odd, so $a^{r/2}$ is not an integer power; the attempt fails and a new a is chosen

Concepts required. An odd order gives no square root $a^{r/2}$ in the integers; the attempt fails.

Relevant formulae. $4, 16, 64 \equiv 1 \pmod{21}$: $r = 3$.

Step-by-step explanation. $4^1, 4^2, 4^3 \equiv 4, 16, 64 = 3 \cdot 21 + 1$, so $r = 3$, odd. The reduction needs r even (then $(a^{r/2} - 1)(a^{r/2} + 1) \equiv 0$). A failed attempt must be reported as a failure, not as a success; one simply repeats with another base ($a = 2$ gives $r = 6$ and the factors 3, 7).

Why the other options are incorrect.

(B) The reduction is not applied for odd r; "4^1" is not $a^{r/2}$ for $r = 3$, and a gcd of 1 is no factor anyway.

(C) $\gcd(4, 21) = 1$; 4 shares no factor with $21 = 3 \cdot 7$.

(D) $4^3 \equiv 1$, so the order is 3, not 6; 6 is the order of $a = 2$ modulo 21.

Key takeaway. Odd order ⇒ retry with a new base; never present it as a success.

Q.45 Single-correct MCQ · Lucky base

For $N = 21$ the randomly chosen base is $a = 6$. What happens?

(A) U_6 is a valid unitary and phase estimation returns $r = 2$

(B) order finding gives $r = 6$ and the factors 3, 7

(C) ✓ $\gcd(6, 21) = 3$ is already a non-trivial factor; no quantum computation is needed

(D) the algorithm fails because a is even

Answer: (C) Detailed solution

Correct answer. (C) $\gcd(6, 21) = 3$ is already a non-trivial factor; no quantum computation is needed

Concepts required. The first step of the algorithm is the classical test $\gcd(a, N)$.

Relevant formulae. $\gcd(6, 21) = 3$, $21/3 = 7$.

Step-by-step explanation. The first classical step is $\gcd(a, N)$. Here $\gcd(6, 21) = 3$, so $21 = 3 \times 7$ immediately (a "lucky" base). Moreover U_6 would not be unitary ($6 \cdot 0 \equiv 6 \cdot 7 \equiv 0 \pmod{21}$), and 6 has no order modulo 21. Even bases are fine in general (e.g. $a = 2$).

Why the other options are incorrect.

(A) U_6 maps $|0\rangle$ and $|7\rangle$ to $|0\rangle$: it is not unitary, so phase estimation cannot be applied.

(B) Order finding is not defined for a not coprime to N: no power of 6 is $\equiv 1 \pmod{21}$.

(D) Even bases are allowed; $a = 2$ works for $N = 21$. The point is the common factor 3.

Key takeaway. Always compute $\gcd(a, N)$ first — sometimes it already solves the problem.

Q.46 Single-correct MCQ · Register sizes for $N = 33$

With the register sizes of `Shor.ipynb` ($n =$ `N.bit_length()`, $t = 2n$), the circuit for $N = 33$ uses

(A) ✓ 18 qubits ($n = 6$, $t = 12$)

(B) 33 qubits

(C) 12 qubits

(D) 15 qubits ($n = 5$, $t = 10$)

Answer: (A) Detailed solution

Correct answer. (A) 18 qubits ($n = 6$, $t = 12$)

Concepts required. Register sizes: $n =$ bit length of N, $t = 2n$ counting qubits, $3n$ in total.

Relevant formulae. $33 = 100001_2$: $n = 6$, $t = 12$, total 18.

Step-by-step explanation. $33 = 100001_2$ has 6 bits, so $n = 6$, $t = 12$ and $t + n = 18$. Five bits only reach $31 < 33$, so $n = 5$ is too small. The count is $3n$ in this construction — far from the N qubits of option 4.

Why the other options are incorrect.

(B) The qubit count grows with $\log_2 N$, not with N; 33 qubits would be 3×11 bits.

(C) 12 is only the counting register $t = 2n$.

(D) $n = 5$ bits only represent $0, \ldots, 31$, which does not contain all residues modulo 33.

Key takeaway. $3n$ qubits in the notebook's construction; better constructions need about $2n$.

Q.47 SINGLE-CORRECT MCQ · $N = 15$, $a = 11$ outcomes

For $N = 15$, $a = 11$ and $t = 8$ counting qubits, the ideal outcome distribution and the result of post-processing $y = 128$ are

(A) peaks at $y \in \{0, 128\}$; $r = 2$, but $11 \equiv -1$, so the factors are trivial

(B) a uniform distribution, because 11 is prime

(C) ✓ peaks at $y \in \{0, 128\}$ with probability $1/2$ each; $128/256 = 1/2 \Rightarrow r = 2$, $\gcd(10, 15) = 5$, $\gcd(12, 15) = 3$

(D) peaks at $y \in \{0, 64, 128, 192\}$; $r = 4$

Answer: (C) Detailed solution

Correct answer. **(C)** peaks at $y \in \{0, 128\}$ with probability $1/2$ each; $128/256 = 1/2 \Rightarrow r = 2$, $\gcd(10, 15) = 5$, $\gcd(12, 15) = 3$
Concepts required. For $r \mid 2^t$ the outcomes are exactly $k2^t/r$, each with probability $1/r$.
Relevant formulae. $11^2 \equiv 1$: $r = 2$, peaks at 0 and 128; $\gcd(11 \mp 1, 15) = 5, 3$.
Step-by-step explanation. $11^2 = 121 = 8 \cdot 15 + 1$, so $r = 2$ and the peaks are at $k \cdot 256/2 = 0, 128$, each with probability $\frac{1}{2}$ (exact, since r divides 2^t). $y = 128$ gives $s/r = 1/2$, $r = 2$; $11 \not\equiv -1 \equiv 14$, and $\gcd(10, 15) = 5$, $\gcd(12, 15) = 3$. $y = 0$ gives no information.
Why the other options are incorrect.
(A) $11 \not\equiv -1 \equiv 14 \pmod{15}$; the gcds 5 and 3 are non-trivial.
(B) Primality of the base is irrelevant; the distribution depends only on the order r.
(D) Four peaks would mean $r = 4$; but $11^2 = 121 \equiv 1$, so $r = 2$.
Key takeaway. The number of peaks is the order; their spacing is $2^t/r$.

Q.48 SINGLE-CORRECT MCQ · Continued fractions, $N = 33$

For $N = 33$, $a = 5$ and $t = 12$, the measurement gives $y = 1229$. The continued-fraction step and the factors are

(A) $1229/4096 \approx 1/3 \Rightarrow r = 3$; the attempt fails

(B) $1229/4096 \approx 3/10 \Rightarrow r = 3$; factors 3 and 11

(C) $1229/4096 \approx 3/10 \Rightarrow r = 10$; $5^5 \equiv -1$, trivial factors

(D) ✓ $1229/4096 \approx 3/10 \Rightarrow r = 10$; $5^5 \equiv 23$, $\gcd(22, 33) = 11$, $\gcd(24, 33) = 3$

Answer: (D) Detailed solution

Correct answer. **(D)** $1229/4096 \approx 3/10 \Rightarrow r = 10$; $5^5 \equiv 23$, $\gcd(22, 33) = 11$, $\gcd(24, 33) = 3$
Concepts required. Continued fractions find the best approximation s/r with $r \leq N$ to $y/2^t$.
Relevant formulae. $1229/4096 = 0.30005 \approx 3/10$; $5^{10} \equiv 1$, $5^5 \equiv 23$; $\gcd(22, 33) = 11$, $\gcd(24, 33) = 3$.
Step-by-step explanation. $1229/4096 = 0.30005$; `limit_denominator(33)` gives $3/10$ (within $\frac{1}{2 \cdot 4096}$). $5^{10} \equiv 1 \pmod{33}$ and no smaller power is 1, so $r = 10$. $5^5 = 3125 = 94 \cdot 33 + 23$; $23 \neq 32$; $\gcd(22, 33) = 11$, $\gcd(24, 33) = 3$. The numerator 3 is s, not the order.
Why the other options are incorrect.
(A) $1/3 = 0.333$ is much farther from 0.30005 than $3/10$; continued fractions give $3/10$.
(B) The numerator 3 is the random index s; the order is the denominator 10.
(C) $5^5 \equiv 23$, and $-1 \equiv 32$; so the factors are non-trivial.
Key takeaway. Read the order from the denominator, then verify $a^r \equiv 1$.

Q.49 Single-correct MCQ · Multiples test

For $N = 33$, $a = 5$, $t = 12$, the outcome is $y = 2048$. The procedure of `candidate_orders` returns

(A) the order 2, because the denominator is always the order

(B) ✓ the fraction $1/2$ and, after testing multiples of 2, the order 10 (since $5^2, 5^4, 5^6, 5^8 \not\equiv 1$, $5^{10} \equiv 1$)

(C) no candidate, because $y = 2^{t-1}$ is never useful

(D) the order 5

Answer: (B) **Detailed solution**

Correct answer. **(B)** the fraction $1/2$ and, after testing multiples of 2, the order 10 (since $5^2, 5^4, 5^6, 5^8 \not\equiv 1$, $5^{10} \equiv 1$)
Concepts required. If $\gcd(s, r) > 1$ the reduced fraction has a denominator that divides r; testing multiples recovers r.
Relevant formulae. $2048/4096 = 1/2 = 5/10$; test $2k$: $5^{10} \equiv 1$ first at $k = 5$.
Step-by-step explanation. $2048/4096 = 1/2$: the measured $s/r = 5/10$ has $\gcd(s, r) = 5$, so it reduces to $1/2$. Testing $2k$: $5^2 \equiv 25$, $5^4 \equiv 31$, $5^6 \equiv 16$, $5^8 \equiv 4$, $5^{10} \equiv 1$ — the first multiple that works is 10. Without the multiples test this outcome would be lost.
Why the other options are incorrect.
(A) The denominator is the order only if $\gcd(s, r) = 1$; here $s = 5$, $r = 10$ share the factor 5.
(C) $y = 2^{t-1}$ is useful whenever r is even: it gives $1/2$, and the multiples test finds r.
(D) $5^5 \equiv 23 \neq 1$, so 5 is not the order; also 5 is not a multiple of the denominator 2.
Key takeaway. The multiples test turns many "partial" outcomes into successes.

Q.50 Single-correct MCQ · Condition $2^t \geq N^2$

The textbook condition $2^t \geq N^2$ guarantees that continued fractions recover s/r from the nearest outcome. For $N = 33$, the smallest such t and the value used by `Shor.ipynb` are

(A) $t = 12$; the notebook uses 11

(B) ✓ $t = 11$; the notebook uses $t = 2n = 12$

(C) $t = 10$; the notebook uses 12

(D) $t = 6$; the notebook uses 12

Answer: (B) **Detailed solution**

Correct answer. **(B)** $t = 11$; the notebook uses $t = 2n = 12$
Concepts required. With $2^t \geq N^2$, the outcome nearest $s2^t/r$ lies within $\frac{1}{2r^2}$ of s/r, so continued fractions find it.
Relevant formulae. $N^2 = 1089$; $2^{10} = 1024 < 1089 \leq 2048 = 2^{11}$.
Step-by-step explanation. $N^2 = 1089$: $2^{10} = 1024 < 1089 \leq 2^{11} = 2048$, so $t = 11$. The notebook uses $t = 2 \cdot$`bit_length` $= 12$, which always satisfies $2^{2n} \geq N^2$ because $N < 2^n$. Then two distinct fractions with denominators $\leq N$ differ by more than $1/N^2 \geq 1/2^t$.
Why the other options are incorrect.
(A) This swaps the two values: the minimum is 11, and the notebook's $t = 2n$ gives 12.
(C) $2^{10} = 1024 < 1089 = N^2$, so $t = 10$ is too small.
(D) $t = 6$ is the work register size n; $2^6 = 64 \ll N^2$.
Key takeaway. $t = 2n$ always satisfies $2^t \geq N^2$, with up to one qubit to spare.

Q.51 Single-correct MCQ · Exact success, notebook default

For the notebook's default ($N = 15$, $a = 7$, $t = 8$) the ideal outcomes are $y \in \{0, 64, 128, 192\}$, each with probability $1/4$. Using the multiples test, the exact probability that one shot yields the factors is

(A) $1/4$ — only $y = 64$ succeeds **(B)** 1

(C) $1/2$ — only $y = 64$ and 192 succeed **(D)** ✓ $3/4$ — only $y = 0$ fails

Answer: (D) Detailed solution

Correct answer. (D) $3/4$ — only $y = 0$ fails

Concepts required. Count the outcomes whose post-processing yields the factors, weighted by their probabilities.

Relevant formulae. $y = 64, 192 \to r = 4$; $y = 128 \to 1/2 \to$ multiples $\to 4$; $y = 0 \to$ nothing: $3/4$.

Step-by-step explanation. $y = 64$, 192 give $1/4$, $3/4$, i.e. $r = 4$ directly; $y = 128$ gives $1/2$, and the multiples test finds 4; $y = 0$ gives no denominator. $7^2 = 49 \equiv 4$, $\gcd(3, 15) = 3$, $\gcd(5, 15) = 5$. Hence $3/4$; the recorded run gave 0.766 (finite-shot fluctuation around 0.75).

Why the other options are incorrect.

(A) $1/4$ counts one successful outcome; three of the four peaks succeed.

(B) Success probability 1 is impossible: the outcome $y = 0$ occurs with probability $1/4$ and carries no information.

(C) $1/2$ forgets that $y = 128$ succeeds through the multiples test.

Key takeaway. Recorded 0.766 vs exact 0.75: the difference is shot noise, not a better algorithm.

Q.52 Single-correct MCQ · Probability $\varphi(r)/r$

Suppose $r = 12$ divides 2^t, so every outcome is exactly $y = s2^t/r$ with s uniform in $\{0, \ldots, 11\}$. *Without* the multiples test, the fraction of shots whose reduced denominator equals r is

(A) ✓ $\varphi(12)/12 = 1/3$ **(B)** $1/2$ **(C)** $11/12$ **(D)** $1/12$

Answer: (A) Detailed solution

Correct answer. (A) $\varphi(12)/12 = 1/3$

Concepts required. The reduced fraction s/r keeps the denominator r iff $\gcd(s, r) = 1$.

Relevant formulae. $P = \varphi(r)/r$; $\varphi(12) = |\{1, 5, 7, 11\}| = 4$, $4/12 = 1/3$.

Step-by-step explanation. The reduced fraction $s/12$ has denominator 12 exactly when $\gcd(s, 12) = 1$: $s \in \{1, 5, 7, 11\}$, i.e. $\varphi(12) = 4$ of 12 values. In general the probability is $\varphi(r)/r = \Omega(1/\log\log r)$, so $O(\log\log N)$ repetitions (or the multiples/lcm trick) suffice.

Why the other options are incorrect.

(B) $1/2$ is the fraction of odd s; $s = 3, 9$ are odd but share the factor 3 with 12.

(C) $11/12$ counts every $s \neq 0$; most of them share a factor with 12.

(D) $1/12$ counts only $s = 1$; $s = 5, 7, 11$ are also coprime to 12.

Key takeaway. $\varphi(r)/r = \Omega(1/\log\log r)$: a few repetitions suffice.

Q.53 SINGLE-CORRECT MCQ · Role of the QFT in Shor

What is the role of the (inverse) quantum Fourier transform *within* Shor's algorithm?

(A) ✓ it is a subroutine of phase estimation that turns the periodic phase pattern on the counting register into peaks near $y = k2^t/r$; it does not factor N by itself

(B) it finds the factors p and q directly from $|N\rangle$

(C) it computes a^x mod N

(D) it is the whole of Shor's algorithm; the remaining steps are optional

Answer: (A) Detailed solution

Correct answer. **(A)** it is a subroutine of phase estimation that turns the periodic phase pattern on the counting register into peaks near $y = k2^t/r$; it does not factor N by itself

Concepts required. The QFT is a component of phase estimation; Shor's algorithm is phase estimation of U_a plus classical processing.

Relevant formulae. $\mathrm{QFT}^{-1} : \frac{1}{\sqrt{2^t}} \sum_x e^{2\pi i x\varphi} |x\rangle \mapsto\approx |2^t\varphi\rangle$.

Step-by-step explanation. Shor's algorithm = classical reduction + order finding (modular exponentiation, QFT, measurement) + classical post-processing. The QFT is one component; the dominant cost of real implementations is the modular exponentiation, not the QFT ($O(n^2)$ gates, or fewer when approximated).

Why the other options are incorrect.

(B) No circuit acts on $|N\rangle$; N enters only through the classical choice of the multipliers.

(C) a^x mod N is computed by the controlled modular multiplications, not by the QFT.

(D) The QFT alone factors nothing; modular exponentiation and the classical steps are essential.

Key takeaway. Never confuse Shor with the QFT: the QFT is one subroutine.

Q.54 SINGLE-CORRECT MCQ · Skipped identity multipliers

`build_shor_kernel` computes $c_j = a^{2^j}$ mod N classically and skips $c_j = 1$. For $N = 15$, $a = 7$, $t = 8$, the number of controlled multiplications actually applied is

(A) 4 **(B)** 1

(C) ✓ 2 ($c_0 = 7$, $c_1 = 4$; $c_j = 1$ for $j \geq 2$) **(D)** 8

Answer: (C) Detailed solution

Correct answer. **(C)** 2 ($c_0 = 7$, $c_1 = 4$; $c_j = 1$ for $j \geq 2$)

Concepts required. The multiplier constants $c_j = a^{2^j}$ mod N are computed classically; $c_j = 1$ is the identity.

Relevant formulae. $7, 49 \equiv 4, 16 \equiv 1, 1, \ldots$: two non-identity multipliers.

Step-by-step explanation. $7^1 = 7$, $7^2 = 49 \equiv 4$, $7^4 \equiv 16 \equiv 1$, and then 1 forever. Only two multipliers remain — matching the recorded gate counts (`ccustom_unitary_constant`: 2; operations `mulmod_7_mod_15`, `mulmod_4_mod_15`). This shortcut uses the known order and is specific to the tiny example.

Why the other options are incorrect.

(A) 4 is the order r, not the number of multipliers.

(B) 1 forgets that $c_1 = 4 \neq 1$ (it appears as `mulmod_4_mod_15`).

(D) 8 would be the count without skipping the identity multipliers; the notebook skips $c_j = 1$.

Key takeaway. Skipping identity multipliers exploits $r \mid 2^j$ — a shortcut available only for toy examples.

Q.55 MATCHING LISTS · Functions of `Shor.ipynb`

Match each function of `Shor.ipynb` (List-I) with what it does (List-II).

List-I		List-II	
(P)	`mulmod_matrix`	**(1)**	brute-force order of a modulo m as a classical reference (feasible only for tiny m)
(Q)	`build_shor_kernel`	**(2)**	continued-fraction denominator of $y/2^t$ and the smallest multiple kr' with $a^{kr'} \equiv 1$
(R)	`candidate_orders`	**(3)**	builds the CUDA-Q circuit: Hadamards, controlled $U_{a^{2^j}}$, inverse QFT, measurement
(S)	`classical_order`	**(4)**	permutation matrix $\|y\rangle \mapsto \|cy \bmod N\rangle$ for $y < N$, identity above

(A) P→3, Q→4, R→2, S→1
(B) ✓ P→4, Q→3, R→2, S→1
(C) P→4, Q→3, R→1, S→2
(D) P→4, Q→2, R→3, S→1

Answer: (B) Detailed solution

Correct answer. (B) P→4, Q→3, R→2, S→1

Concepts required. The notebook separates matrix construction, circuit construction, post-processing and classical checks.

Relevant formulae. `mulmod_matrix` → `cudaq.register_operation` → `build_shor_kernel` → `cudaq.sample` → `candidate_orders`.

Step-by-step explanation. `mulmod_matrix` builds the unitary (4); `build_shor_kernel` assembles the circuit with `cudaq.register_operation` gates (3); `candidate_orders` is the classical post-processing (2); `classical_order` is the brute-force check (1). Hence P→4, Q→3, R→2, S→1.

Why the other options are incorrect.

(A) This swaps the matrix and the kernel builder: `mulmod_matrix` returns a NumPy permutation matrix, not a circuit.

(C) This swaps post-processing and the classical reference: `candidate_orders` uses y and continued fractions; `classical_order` uses brute force.

(D) This swaps the kernel builder and the post-processing: `build_shor_kernel` builds the circuit and never sees y.

Key takeaway. Know which part of the code is quantum (the kernel) and which is classical (everything around it).

Q.56 MATCHING LISTS · Outcomes to fractions

For $N = 33$, $a = 5$ ($r = 10$) and $t = 12$, match each outcome y (List-I) with `Fraction(y, 4096).limit_denominator(33)` (List-II).

List-I		List-II	
(P)	410	**(1)**	2/5
(Q)	819	**(2)**	7/10
(R)	1638	**(3)**	1/10
(S)	2867	**(4)**	1/5

(A) P→3, Q→4, R→2, S→1
(B) P→3, Q→1, R→4, S→2
(C) P→4, Q→3, R→1, S→2
(D) ✓ P→3, Q→4, R→1, S→2

Answer: (D) Detailed solution

Correct answer. (D) P→3, Q→4, R→1, S→2
Concepts required. `limit_denominator(N)` returns the closest fraction with denominator at most N.
Relevant formulae. $410/4096 = 0.1001$, $819/4096 = 0.19995$, $1638/4096 = 0.39990$, $2867/4096 = 0.69995$.
Step-by-step explanation. $410/4096 = 0.1001 \approx 1/10$; $819/4096 = 0.19995 \approx 1/5$; $1638/4096 = 0.3999 \approx 2/5$; $2867/4096 = 0.69995 \approx 7/10$. Only 410 and 2867 give the denominator 10 directly ($s = 1, 7$ coprime to 10); 819 and 1638 give 5, and the multiples test then finds 10.
Why the other options are incorrect.

(A) This exchanges 2/5 and 7/10: $1638/4096 \approx 0.4$ and $2867/4096 \approx 0.7$.

(B) This assigns 1/10 to 1638; but $1638/4096 \approx 0.4 = 2/5$.

(C) This exchanges 1/10 and 1/5: $410/4096 \approx 0.1$ and $819/4096 \approx 0.2$.

Key takeaway. Only fractions with $\gcd(s, r) = 1$ give r directly; others need the multiples test.

Q.57 Assertion–Reason · Prime powers

Assertion (A): Shor's order-finding reduction is the method of choice for factoring $N = 81 = 3^4$.
Reason (R): For an odd prime power $N = p^k$, the only square roots of 1 modulo N are ± 1.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) ✓ A is false, but R is true

Answer: (D) Detailed solution

Correct answer. (D) A is false, but R is true
Concepts required. Prime powers are excluded from the quantum reduction and handled classically.
Relevant formulae. $\mathbb{Z}_{p^k}^*$ cyclic $\Rightarrow x^2 \equiv 1 \iff x \equiv \pm 1$; $81 = 3^4$ found by integer roots.
Step-by-step explanation. **A is false**: prime powers are detected classically (integer k-th roots) — `Shor.ipynb` does so with `prime_power` before any quantum step. **R is true**: $\mathbb{Z}_{p^k}^*$ is cyclic, so $x^2 \equiv 1$ has only the solutions ± 1; hence $a^{r/2}$ would always be ± 1 and $\gcd(a^{r/2} \pm 1, N)$ would be trivial. Answer: (D).
Why the other options are incorrect.

(A) is wrong because A is false: prime powers are detected classically (`prime_power` in the notebook).

(B) is wrong because A is false, so it cannot be a true assertion with an unrelated reason.

(C) is wrong because A is false and R is true, the opposite of (C).

Key takeaway. Shor needs at least two distinct prime factors; check prime powers classically first.

Q.58 Statement I / II · Uninformative outcomes

Statement I: A measured value $y = 0$ means that the order is $r = 1$.

Statement II: When r does not divide 2^t, every measured y yields the exact order through continued fractions.

(A) Both Statement I and Statement II are true

(B) ✓ Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (B) Detailed solution

Correct answer. **(B)** Both Statement I and Statement II are false

Concepts required. Measurement outcomes are random; only some of them reveal the order directly.

Relevant formulae. $y = 0 \leftrightarrow s = 0$ (always present); success requires $y \approx s2^t/r$ with $\gcd(s, r) = 1$.

Step-by-step explanation. **I is false**: $y = 0$ is the peak $s = 0$, present for every r; it carries no information (the notebook returns no candidate). **II is false**: only outcomes close to $s2^t/r$ with $\gcd(s, r) = 1$ give r directly; others give a divisor of r or nothing, and the probability mass is spread over neighbouring y. Answer: (B).

Why the other options are incorrect.

(A) is wrong because both statements are false.

(C) is wrong because Statement I is false: $y = 0$ occurs for every order and gives no denominator.

(D) is wrong because Statement II is false: for $N = 21$, $a = 2$ the outcome $y = 512$ gives $1/2$, and outcomes between peaks give nothing useful.

Key takeaway. Shor's algorithm is probabilistic; failures are expected and handled by repetition.

Paragraph for Questions 59 and 60

Shor's algorithm is applied to $N = 39$ with base $a = 2$, using the register sizes of `Shor.ipynb`: the work register has $n =$ `N.bit_length()` $= 6$ qubits and the counting register $t = 2n = 12$ qubits, so the peaks of the outcome distribution lie near $y \approx k \cdot 4096/r$. A measured value y is post-processed with `Fraction(y, 2**t).limit_denominator(N)`; the denominator r' of the result, or the smallest multiple kr' with $a^{kr'} \equiv 1 \pmod N$, is taken as the order. The powers of 2 modulo 39 are $2, 4, 8, 16, 32, 25, 11, 22, 5, 10, 20, 1$.

Q.59 Paragraph based · $N = 39$: order and factors

The order r and the factors $\gcd(2^{r/2} - 1, 39)$, $\gcd(2^{r/2} + 1, 39)$ are

(A) ✓ $r = 12$; 3 and 13

(B) $r = 6$; 3 and 13

(C) $r = 12$; 7 and 13

(D) $r = 12$; 1 and 39

Answer: (A) Detailed solution

Correct answer. **(A)** $r = 12$; 3 and 13

Concepts required. The order is the position of the first 1 in the list of powers.

Relevant formulae. $2^{12} \equiv 1 \pmod{39}$, $2^6 \equiv 25$; $\gcd(24, 39) = 3$, $\gcd(26, 39) = 13$.

Step-by-step explanation. From the list, $2^{12} \equiv 1$ is the first 1: $r = 12$, even. $2^6 \equiv 25 \not\equiv 38$. $\gcd(24, 39) = 3$ and $\gcd(26, 39) = 13$; $3 \times 13 = 39$. $r = 6$ would require $2^6 \equiv 1$, but $2^6 \equiv 25$.

Why the other options are incorrect.

(B) $2^6 \equiv 25 \neq 1$, so $r \neq 6$ (and the factors would have to come from 2^3).

(C) 7 does not divide $39 = 3 \cdot 13$.

(D) $25 \not\equiv -1 \equiv 38$, so the gcds are non-trivial.

Key takeaway. $N = 39$, $a = 2$: $r = 12$, $39 = 3 \times 13$.

Q.60 PARAGRAPH BASED · $N = 39$: continued fraction

The outcome $y = 1707$ is measured. The continued-fraction step gives

(A) $5/12$, so $r = 5$

(B) $1/2$, so the multiples test is needed

(C) ✓ $5/12$, so $r = 12$ directly

(D) $2/5$, so $r = 5$

Answer: (C) Detailed solution

Correct answer. (C) $5/12$, so $r = 12$ directly

Concepts required. A continued-fraction convergent within $\frac{1}{2r^2}$ of $y/2^t$ is the reduced form of s/r.

Relevant formulae. $|1707/4096 - 5/12| = 8.1 \times 10^{-5} < 1/(2 \cdot 39^2) = 3.3 \times 10^{-4}$.

Step-by-step explanation. $1707/4096 = 0.41675$, and $5/12 = 0.41667$; $|1707/4096 - 5/12| = 8.1 \times 10^{-5} < 1/(2 \cdot 39^2)$, so `limit_denominator(39)` returns $5/12$. $\gcd(5, 12) = 1$, so the denominator is the order 12. The numerator 5 is the random index s, not the order.

Why the other options are incorrect.

(A) The numerator 5 is the random index s; the order is the denominator 12.

(B) $1/2 = 0.5$ is far from 0.41675; that fraction belongs to $y = 2048$.

(D) $2/5 = 0.4$ differs by 0.017, while $5/12$ differs by only 8×10^{-5}.

Key takeaway. Denominator = order when $\gcd(s, r) = 1$; always verify $a^r \equiv 1$.

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SECTION D · Advanced Shor's Algorithm and Implementation Q.61–80

Q.61 Single-correct MCQ · Beauregard qubit count

Beauregard's circuit for Shor's algorithm uses $2n + 3$ logical qubits. For RSA-2048 ($n = 2048$) this is

(A) 6147 **(B)** ✓ 4099 **(C)** 4096 **(D)** 2051

Answer: (B) Detailed solution

Correct answer. (B) 4099
Concepts required. Logical-qubit counts of Shor circuits are linear in the bit length n.
Relevant formulae. Beauregard: $2n + 3$; $n = 2048$: 4099.
Step-by-step explanation. $2 \times 2048 + 3 = 4099$. $6147 = 3 \times 2048 + 3$ resembles the $3n$ register count of `Shor.ipynb` (6144). These are *logical* (error-corrected) qubits; the physical count is orders of magnitude larger.
Why the other options are incorrect.
(A) $6147 = 3n + 3$; the $3n$ register count belongs to the notebook's construction, not Beauregard's.
(C) $4096 = 2n$ forgets the three extra qubits (carry/ancilla and the recycled control).
(D) $2051 = n + 3$ omits one of the two n-qubit registers (work register and adder register).
Key takeaway. Thousands of logical qubits — each made of hundreds to thousands of physical qubits.

Q.62 Single-correct MCQ · Published RSA-2048 estimates

Which pair of published resource estimates for factoring RSA-2048 is correct?

(A) Gidney (2025): about 20 billion qubits in about 8 hours
(B) both estimates state that RSA-2048 was factored on existing hardware
(C) ✓ Gidney and Ekerå (2019): about 20 million noisy qubits in about 8 hours; Gidney (2025): fewer than one million noisy qubits in less than a week
(D) Gidney and Ekerå (2019): 4099 physical qubits in one second

Answer: (C) Detailed solution

Correct answer. (C) Gidney and Ekerå (2019): about 20 million noisy qubits in about 8 hours; Gidney (2025): fewer than one million noisy qubits in less than a week
Concepts required. Published resource estimates are model-based projections, not experimental results.
Relevant formulae. GE2019: $\approx 2 \times 10^7$ physical qubits, ≈ 8 h; Gidney 2025: $< 10^6$ physical qubits, < 1 week.
Step-by-step explanation. Both are *estimates* under explicit assumptions (surface code, physical error rate 10^{-3}, μs cycle times), not experiments. Gidney–Ekerå 2019: ≈ 20 million physical qubits, ≈ 8 hours. Gidney 2025 (approximate residue arithmetic, yoked surface codes, magic-state cultivation): < 1 million qubits, < 1 week. No cryptographically relevant number has been factored by Shor's algorithm.
Why the other options are incorrect.
(A) The 2025 estimate *lowered* the qubit count to under one million; 20 billion is fabricated.
(B) No resource estimate claims an actual factorisation of RSA-2048; none has been performed.
(D) 4099 is Beauregard's *logical* count; GE2019 required about 20 million physical qubits and hours, not seconds.

Key takeaway. Distinguish estimates from demonstrations: RSA-2048 remains unfactored.

Q.63 Single-correct MCQ · $O(n^3)$ scaling

With schoolbook arithmetic, modular exponentiation costs $O(n^3)$ gates. Doubling the bit length n of N multiplies the gate count by about

(A) 2 **(B)** 2^n **(C)** ✓ 8 **(D)** 4

Answer: (C) Detailed solution

Correct answer. **(C)** 8
Concepts required. Polynomial scaling: $O(n^3)$ means doubling n multiplies the cost by 2^3.
Relevant formulae. $(2n)^3/n^3 = 8$.
Step-by-step explanation. $(2n)^3 = 8n^3$. The cost is polynomial: $t = 2n$ controlled multiplications, each $O(n^2)$ gates. Classical sub-exponential algorithms (GNFS) grow much faster, which is the source of Shor's advantage — but only on a fault-tolerant machine. 2^n is the exponential cost of trial division.
Why the other options are incorrect.
(A) 2 would be linear scaling, $O(n)$.
(B) 2^n is exponential growth, the cost of brute-force methods, not of Shor's algorithm.
(D) 4 would be quadratic scaling, $O(n^2)$ — e.g. the QFT alone.
Key takeaway. Shor is polynomial ($O(n^3)$ or better), GNFS sub-exponential — the gap grows with n.

Q.64 Single-correct MCQ · Semiclassical QFT

In the "semiclassical" (one-control-qubit) implementation of the inverse QFT used by Beauregard, the $t = 2n$ counting qubits are replaced by

(A) $2n$ ancilla qubits that are never measured
(B) n qubits, because the QFT needs only half the bits
(C) no qubits: the QFT is computed classically before the circuit
(D) ✓ a single qubit that is measured and reset $2n$ times; earlier measurement results classically control the phase corrections

Answer: (D) Detailed solution

Correct answer. **(D)** a single qubit that is measured and reset $2n$ times; earlier measurement results classically control the phase corrections
Concepts required. The inverse QFT followed by measurement can be performed one qubit at a time with classical feed-forward.
Relevant formulae. Bit y_k measured → rotations $R_1(-\pi \sum_j y_j 2^{-(j-k)})$ on the next qubit; $3n \to 2n+3$ qubits.
Step-by-step explanation. The inverse QFT followed by measurement can be done bit by bit (Griffiths–Niu): measure, then apply classically controlled phases to the next qubit. One control qubit is recycled, so the count drops from $3n$ to about $2n+3$ (n work, $n+2$ ancilla for the Fourier-space adder, 1 control). The measurement statistics are unchanged.
Why the other options are incorrect.
(A) Ancilla qubits that are never measured would not reduce the qubit count; the point is to recycle one qubit.
(B) The QFT needs all $2n$ bits of precision; they are measured sequentially on one qubit, not reduced to n qubits.

(C) The QFT acts on quantum amplitudes, which cannot be computed classically in advance.

Key takeaway. Measurement plus classical control trades qubits for mid-circuit measurements.

Q.65 SINGLE-CORRECT MCQ · Approximate QFT

For $N = 21$, $a = 2$, $t = 10$, an approximate inverse QFT keeps only controlled phases between qubits at distance $k - j \le m$. With $m = 2$ the number of controlled phases and the (re-executed, exact) success probability are

(A) ✓ 17 phases; 0.689 (full QFT: 45 phases, 0.817)

(B) 45 phases; 0.817

(C) 17 phases; 0.817 — truncation never changes the result

(D) 2 phases; 0.689

Answer: (A) Detailed solution

Correct answer. **(A)** 17 phases; 0.689 (full QFT: 45 phases, 0.817)

Concepts required. The approximate QFT drops small-angle controlled phases; it reduces the gate count at a small cost in accuracy.

Relevant formulae. Pairs at distance ≤ 2: $9 + 8 = 17$; full $\binom{10}{2} = 45$.

Step-by-step explanation. Distance 1: 9 pairs, distance 2: 8 pairs, total 17 (full QFT: $\binom{10}{2} = 45$). The re-executed simulation gave success 0.689 for $m = 2$, 0.787 for $m = 3$, 0.811 for $m = 4$ and 0.817 for the full QFT. Very small cut-offs lose accuracy; $m \approx \log_2 n$ plus a few suffices asymptotically.

Why the other options are incorrect.

(B) 45 phases and 0.817 describe the full QFT, not the truncation $m = 2$.

(C) Truncation changes the amplitudes; the re-executed simulation shows a drop from 0.817 to 0.689.

(D) $m = 2$ is the maximum distance, not the number of gates; each distance contributes $t - d$ pairs.

Key takeaway. Moderate cut-offs ($m \approx 4$) lose little; aggressive ones lose a lot.

Q.66 SINGLE-CORRECT MCQ · Controlled phases in the QFT

The full inverse QFT on $t = 8$ counting qubits (notebook default for $N = 15$) contains how many controlled phase gates?

(A) 8 **(B)** ✓ 28 **(C)** 64 **(D)** 56

Answer: (B) Detailed solution

Correct answer. **(B)** 28

Concepts required. The QFT on t qubits uses one controlled phase per qubit pair.

Relevant formulae. $t(t-1)/2 = 28$ for $t = 8$.

Step-by-step explanation. One for each pair $j < k$: $\binom{8}{2} = \frac{8 \cdot 7}{2} = 28$ — the recorded gate count `cr1: 28`. In general $t(t-1)/2 = O(t^2)$, which for large t is dwarfed by the $O(n^3)$ cost of the modular exponentiation.

Why the other options are incorrect.

(A) 8 is the number of Hadamards in the inverse QFT, not of controlled phases.

(C) $64 = t^2$ counts ordered pairs including $j = k$.

(D) $56 = t(t-1)$ counts each unordered pair twice.

Key takeaway. QFT cost is $O(t^2)$ — small compared with modular exponentiation.

Q.67 SINGLE-CORRECT MCQ · Recorded gate counts

The recorded gate counts of `Shor.ipynb` for $N = 15$ are `x: 1, ccustom_unitary_constant: 2, swap: 4, h: 16, cr1: 28` (and `mz: 8`). Why are there 16 Hadamard gates?

(A) ✓ 8 create the superposition on the counting qubits and 8 belong to the inverse QFT

(B) $16 = 2^4$ is the dimension of the work register

(C) each controlled multiplication contains 8 Hadamards

(D) Hadamards are inserted before every measurement of the work register

Answer: (A) Detailed solution

Correct answer. **(A)** 8 create the superposition on the counting qubits and 8 belong to the inverse QFT

Concepts required. Hadamards appear at the start (superposition) and in the inverse QFT (one per qubit).

Relevant formulae. $8 + 8 = 16$; total $1 + 2 + 4 + 16 + 28 = 51$ gates without `mz`.

Step-by-step explanation. $t = 8$ Hadamards start the algorithm, and the inverse QFT applies one H per qubit (8 more). The multipliers are single custom gates (no internal H), and the work register is not measured. Total gates without `mz`: $1 + 2 + 4 + 16 + 28 = 51$, depth 20.

Why the other options are incorrect.

(B) The coincidence $16 = 2^4$ is irrelevant; the count is $t + t$.

(C) The multipliers are single `ccustom_unitary_constant` gates; no Hadamards are inside them.

(D) The work register is not measured in the notebook; only `mz(count)` appears.

Key takeaway. Gate counts should be traced back to the circuit's structure.

Q.68 SINGLE-CORRECT MCQ · No hardware execution

Why does `Shor.ipynb` refuse to run on hardware targets?

(A) CUDA-Q has no hardware targets

(B) the inverse QFT cannot be executed on hardware

(C) Shor's algorithm is mathematically impossible on hardware

(D) ✓ its modular multipliers are exact custom unitaries (permutation matrices) registered with `cudaq.register_operation`; hardware would need them decomposed into native gates, which the notebook does not do

Answer: (D) Detailed solution

Correct answer. **(D)** its modular multipliers are exact custom unitaries (permutation matrices) registered with `cudaq.register_operation`; hardware would need them decomposed into native gates, which the notebook does not do

Concepts required. Custom matrix gates are a simulator convenience; hardware executes only native gate sets.

Relevant formulae. `cudaq.register_operation(name, matrix)`; hardware → `ValueError` in the notebook.

Step-by-step explanation. The notebook raises a `ValueError` for hardware and points to the resource estimates. A dense $2^n \times 2^n$ matrix gate is fine for a simulator but has no native implementation; compiling it costs $O(n^2)$–$O(n^3)$ gates per multiplier. The inverse QFT itself uses standard gates.

Why the other options are incorrect.

(A) CUDA-Q supports several hardware providers; the restriction is the notebook's custom gates.

(B) The inverse QFT uses H, controlled phases and SWAPs, which compile to native gates.

(C) Shor's algorithm is a well-defined circuit; small instances have run on hardware with compiled (often simplified) circuits.

Key takeaway. Running on hardware requires a gate-level multiplier — the hard engineering part.

Q.69 SINGLE-CORRECT MCQ · CNOTs of a controlled multiplier

In the gate-level circuit, "multiply by 7 modulo 15" is built as three left rotations of the 4 work bits (each 3 SWAPs) followed by X on all four bits. Its controlled version, with controlled SWAPs of 8 CNOTs and controlled-X = CNOT, uses how many CNOTs?

(A) 9 **(B)** 40 **(C)** ✓ 76 **(D)** 72

Answer: (C) Detailed solution

Correct answer. **(C)** 76

Concepts required. Count CNOTs gate by gate: controlled SWAP = 8, controlled X = 1.

Relevant formulae. $9 \times 8 + 4 = 76$; whole circuit: $76 + 48 + 3 = 127$.

Step-by-step explanation. 9 controlled SWAPs $\times$ 8 = 72 CNOTs, plus 4 CNOTs for the controlled X gates: 76. (Multiplication by 7 is valid on $y = 1, \ldots, 14$: $7y \bmod 15 = 15 - (y \cdot 2^3 \bmod 15)$.) With the controlled $\times 4$ (6 controlled SWAPs = 48) and the QFT SWAP (3) the circuit has 127 CNOTs.

Why the other options are incorrect.

(A) 9 counts controlled SWAPs, not CNOTs.

(B) 40 would use 4 CNOTs per controlled SWAP; a Toffoli alone needs 6.

(D) 72 forgets the four controlled X gates.

Key takeaway. Even multiplying a 4-bit number by 7 costs 76 CNOTs when controlled.

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Q.70 SINGLE-CORRECT MCQ · Random baseline

In the gate-level circuit ($t = 3$, $a = 7$, $N = 15$) the counting outcome becomes uniformly random in the fully depolarized limit. The success probability with the notebook's post-processing is then

(A) 1/8

(B) ✓ 3/8: $y = 2, 4, 6$ give $r = 4$ and the factors; $y = 0$ gives nothing; odd y give $r' = 8$, and $7^4 \equiv 1$ gives only trivial factors

(C) 3/4, the same as without noise

(D) 0

Answer: (B) Detailed solution

Correct answer. **(B)** 3/8: $y = 2, 4, 6$ give $r = 4$ and the factors; $y = 0$ gives nothing; odd y give $r' = 8$, and $7^4 \equiv 1$ gives only trivial factors

Concepts required. A uniformly random outcome is the zero-information baseline for any noisy run.

Relevant formulae. Successful $y \in \{2, 4, 6\}$: $3/8 = 0.375$; odd $y \to 8$, $7^4 \equiv 1 \to$ trivial gcds.

Step-by-step explanation. $y/8$: $2/8 = 1/4$, $6/8 = 3/4$ give 4; $4/8 = 1/2$ gives 4 via multiples. For odd y the denominator 8 satisfies $7^8 \equiv 1$, but $7^4 \equiv 1$, so $\gcd(0, 15) = 15$, $\gcd(2, 15) = 1$ — trivial. Hence $3/8 = 0.375$: a noisy run can still "succeed" by luck, which is why a success rate must be compared with this random baseline.

Why the other options are incorrect.

(A) 1/8 counts only one of the three successful outcomes.

(C) 3/4 is the noiseless success probability; full depolarization destroys the peaks.

(D) 0 ignores that random outcomes can still give $r = 4$ by chance.

Key takeaway. Compare noisy success rates with the random baseline, never with zero.

Q.71 Single-correct MCQ · Memory for $N = 1023$

Using the notebook's $3n$-qubit construction, the state-vector memory (16 bytes per amplitude) for $N = 1023$ is about

(A) 1.1 TB (36 qubits)
(B) 16.4 kB
(C) 268 MB (24 qubits)
(D) ✓ 17.2 GB (30 qubits)

Answer: (D) Detailed solution

Correct answer. (D) 17.2 GB (30 qubits)
Concepts required. Memory of the full state vector: $16 \cdot 2^{3n}$ bytes in the notebook's construction.
Relevant formulae. 1023: $n = 10$, $q = 30$, $M = 16 \cdot 2^{30} = 17.2$ GB.
Step-by-step explanation. 1023 has 10 bits, so $3n = 30$ qubits: $16 \times 2^{30} = 17.18 \times 10^9$ bytes. This matches the notebook's reverse estimate that 80 GB allows at most 32 qubits, i.e. $N \leq 1023$. $N = 1024$ would need 11 bits and 33 qubits (137 GB).
Why the other options are incorrect.
(A) 1.1 TB corresponds to 36 qubits, i.e. a 12-bit N.
(B) 16.4 kB is $16 \cdot 2^{10}$: it uses n qubits instead of $3n$.
(C) 268 MB corresponds to 24 qubits, i.e. an 8-bit N such as 143.
Key takeaway. Simulation memory explodes: a few more bits of N exhaust any computer.

Q.72 Single-correct MCQ · Toy RSA

Toy RSA: $p = 5$, $q = 11$, $e = 7$. The private exponent d and the ciphertext of $m = 2$ are

(A) ✓ $d = 23$; $c = 18$
(B) $d = 3$; $c = 18$
(C) $d = 23$; $c = 14$
(D) $d = 7$; $c = 2$

Answer: (A) Detailed solution

Correct answer. (A) $d = 23$; $c = 18$
Concepts required. RSA: $ed \equiv 1 \pmod{\varphi(N)}$, $c = m^e \bmod N$, $m = c^d \bmod N$.
Relevant formulae. $\varphi(55) = 40$, $7 \cdot 23 = 161 \equiv 1$; $2^7 = 128 \equiv 18 \pmod{55}$.
Step-by-step explanation. $\varphi(55) = 4 \cdot 10 = 40$; $7d \equiv 1 \pmod{40}$: $7 \cdot 23 = 161 = 4 \cdot 40 + 1$, so $d = 23$. $c = 2^7 \bmod 55 = 128 - 110 = 18$; check $18^{23} \bmod 55 = 2$. Shor's algorithm breaks RSA by factoring $N = 55$, which reveals $\varphi(N)$ and hence d.
Why the other options are incorrect.
(B) $7 \cdot 3 = 21 \not\equiv 1 \pmod{40}$, so $d \neq 3$.
(C) $c = 2^7 \bmod 55 = 18$, not 14.
(D) $d = e$ would need $49 \equiv 1 \pmod{40}$; and encryption does change the message.
Key takeaway. Factoring N gives $\varphi(N)$ and hence d: that is the threat.

Q.73 Single-correct MCQ · Post-quantum standards

Which standard specifies NIST's lattice-based key-encapsulation mechanism intended to replace RSA/ECDH key exchange?

(A) ✓ FIPS 203 (ML-KEM)
(B) SP 800-90A (random bit generators)
(C) FIPS 197 (AES)
(D) FIPS 186 (DSA/ECDSA)

Answer: (A) Detailed solution

Correct answer. **(A)** FIPS 203 (ML-KEM)
Concepts required. NIST's post-quantum standards: FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), FIPS 205 (SLH-DSA), 2024.
Relevant formulae. ML-KEM: module learning with errors; no known efficient quantum algorithm.
Step-by-step explanation. NIST published FIPS 203 (ML-KEM, from CRYSTALS-Kyber), FIPS 204 (ML-DSA, from Dilithium) and FIPS 205 (SLH-DSA, from SPHINCS+) in August 2024. AES (FIPS 197) is symmetric and not broken by Shor; FIPS 186 contains the signature schemes Shor's algorithm does break.
Why the other options are incorrect.
(B) SP 800-90A specifies random bit generators, not key exchange.
(C) FIPS 197 is AES, a symmetric cipher; it is not a key-encapsulation mechanism.
(D) FIPS 186 specifies DSA/ECDSA/RSA signatures — schemes broken by Shor's algorithm.
Key takeaway. Replace RSA/ECC key exchange by ML-KEM (often in hybrid mode).

Q.74 Single-correct MCQ · Impact on RSA, ECC, AES

Which statement about the cryptographic impact of a large fault-tolerant quantum computer is correct?

(A) hash functions such as SHA-256 are broken by Shor's algorithm
(B) only RSA is affected; elliptic-curve cryptography is immune
(C) ✓ Shor's algorithm breaks RSA and elliptic-curve cryptography; AES is affected only by Grover's quadratic speed-up, so AES-256 remains adequate
(D) Shor's algorithm breaks AES-256 in polynomial time

Answer: (C) Detailed solution

Correct answer. **(C)** Shor's algorithm breaks RSA and elliptic-curve cryptography; AES is affected only by Grover's quadratic speed-up, so AES-256 remains adequate
Concepts required. Shor solves the hidden subgroup problem in abelian groups: factoring and discrete logarithms, including on elliptic curves.
Relevant formulae. Grover: $O(\sqrt{2^k})$ for a k-bit key: AES-256 $\to\approx 2^{128}$ work.
Step-by-step explanation. Shor's period finding solves factoring and discrete logarithms in any abelian group, including elliptic-curve groups (ECDH, ECDSA). Symmetric ciphers and hash functions have no such structure; Grover search gives at most a square-root speed-up, mitigated by doubling key length.
Why the other options are incorrect.
(A) Hash functions are affected only by generic (Grover-type) speed-ups, not by Shor.
(B) ECC relies on discrete logarithms, which Shor's algorithm also solves (with fewer qubits than RSA of equal security).
(D) Shor's algorithm needs algebraic periodicity; AES has none, and Grover gives only a square-root speed-up.

Key takeaway. Public-key cryptography falls; symmetric cryptography needs at most larger keys.

Q.75 MATCHING LISTS · Fault-tolerance terms

Match each fault-tolerance term (List-I) with its meaning (List-II).

List-I		List-II	
(P)	logical qubit	**(1)**	minimum number of physical errors that can cause an undetected logical error
(Q)	physical qubit	**(2)**	a special resource state that enables non-Clifford gates such as T or Toffoli in the surface code
(R)	code distance d	**(3)**	an error-corrected qubit encoded in many noisy qubits
(S)	magic state	**(4)**	a noisy hardware qubit with error rates around 10^{-3}

(A) P→3, Q→4, R→2, S→1
(B) P→3, Q→1, R→4, S→2
(C) P→4, Q→3, R→1, S→2
(D) ✓ P→3, Q→4, R→1, S→2

Answer: (D) Detailed solution

Correct answer. **(D)** P→3, Q→4, R→1, S→2
Concepts required. Fault tolerance encodes logical qubits in many physical qubits and supplies non-Clifford gates via magic states.
Relevant formulae. Surface code: $\approx 2d^2$ physical qubits per logical qubit; logical error $\sim (p/p_{\text{th}})^{(d+1)/2}$.
Step-by-step explanation. Logical qubit: encoded, error-corrected (3); physical qubit: the noisy hardware element (4); code distance: the number of errors needed for an undetected logical error — a surface-code patch uses about $2d^2$ physical qubits (1); magic states: distilled or cultivated resources for T/Toffoli (2). Shor's arithmetic is dominated by Toffoli gates, so magic-state production dominates its cost.
Why the other options are incorrect.

(A) This swaps code distance and magic states; the distance is a number of errors, magic states are resources.

(B) This gives the physical qubit the meaning of code distance; a physical qubit is the noisy hardware element.

(C) This swaps logical and physical qubits; the logical qubit is the encoded one.

Key takeaway. Shor's cost on real hardware is dominated by error correction and magic states.

Q.76 MATCHING LISTS · Qubit counts of constructions

Match each construction for an n-bit N (List-I) with its logical-qubit count (List-II).

List-I		List-II	
(P)	`Shor.ipynb` ($t = 2n$ counting + n work)	**(1)**	$n + 1$
(Q)	Beauregard (2003)	**(2)**	$3n + 0.002\, n \log_2 n$
(R)	Gidney–Ekerå (2019)	**(3)**	$3n$
(S)	one recycled counting qubit + n work qubits with exact (oracle) multipliers	**(4)**	$2n + 3$

(A) P→3, Q→4, R→1, S→2
(B) P→3, Q→2, R→4, S→1
(C) ✓ P→3, Q→4, R→2, S→1
(D) P→4, Q→3, R→2, S→1

Answer: (C) Detailed solution

Correct answer. (C) P→3, Q→4, R→2, S→1
Concepts required. Different constructions trade qubits for gates and measurements.
Relevant formulae. $3n$ (notebook), $2n + 3$ (Beauregard), $3n + 0.002n \log_2 n$ (GE2019), $n + 1$ (idealised semiclassical).
Step-by-step explanation. Notebook: $2n + n = 3n$ (3). Beauregard: $2n + 3$ (4). Gidney–Ekerå: $3n + 0.002n \log_2 n$, about 6190 for $n = 2048$ (2). With the semiclassical QFT and idealised matrix multipliers one needs only $n + 1$ (1) — but real multipliers need ancilla qubits.
Why the other options are incorrect.
(A) This swaps Gidney–Ekerå and the idealised semiclassical count.
(B) This swaps Beauregard and Gidney–Ekerå; GE2019 uses slightly more than $3n$ qubits but far fewer gates.
(D) This swaps the notebook and Beauregard: the notebook uses $t + n = 3n$.
Key takeaway. Qubit counts alone do not decide feasibility; gate counts and error rates matter too.

Q.77 ASSERTION–REASON · Harvest now, decrypt later

Assertion (A): Migrating to post-quantum cryptography is urgent even though no current quantum computer can run Shor's algorithm at cryptographic sizes.

Reason (R): Shor's algorithm also solves the discrete-logarithm problem.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) ✓ Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Answer: (B) Detailed solution

Correct answer. (B) Both A and R are true, but R is *not* the correct explanation of A
Concepts required. Data with long confidentiality lifetimes is already at risk from future quantum computers.
Relevant formulae. Urgent if (data lifetime) + (migration time) > (time to a cryptographically relevant quantum computer).
Step-by-step explanation. **A is true**: encrypted data recorded today can be decrypted later

("harvest now, decrypt later"), and migrations take years. **R is true**: period finding solves discrete logarithms, so Diffie–Hellman and ECC fall too. But R does not explain the *urgency* — that comes from data lifetime and migration time. Answer: (B).

Why the other options are incorrect.

(A) is wrong because R, although true, does not explain the urgency; that comes from harvest-now-decrypt-later and slow migration.

(C) is wrong because R is true: Shor's algorithm solves discrete logarithms.

(D) is wrong because A is true.

Key takeaway. The threat is about the future decryption of today's recorded data.

Q.78 STATEMENT I / II · Qubits and repeated squaring

Statement I: In the notebook's construction, a 32-bit N needs 96 qubits.

Statement II: Repeated squaring lets $a^x \bmod N$ be implemented with t controlled multiplications by the classically precomputed constants $a^{2^j} \bmod N$.

(A) ✓ Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Statement I and Statement II are true

Concepts required. Qubit count $3n$ and the repeated-squaring structure of modular exponentiation.

Relevant formulae. $3 \cdot 32 = 96$; $a^x = \prod_{j=0}^{t-1} (a^{2^j})^{x_j}$.

Step-by-step explanation. **I is true**: $3n = 96$ (the notebook's reverse estimate: 100 qubits allow 33 bits). **II is true**: $a^x = \prod_j (a^{2^j})^{x_j}$, so counting qubit j controls the multiplication by $a^{2^j} \bmod N$, computed classically. Answer: (A).

Why the other options are incorrect.

(B) is wrong because both statements are true.

(C) is wrong because Statement II is true: the controlled multipliers by a^{2^j} are exactly the notebook's construction.

(D) is wrong because Statement I is true: $t + n = 64 + 32 = 96$.

Key takeaway. t controlled multiplications with classically precomputed constants implement $a^x \bmod N$.

Paragraph for Questions 79 and 80

A gate-level version of the order-finding circuit for $N = 15$, $a = 7$ uses $t = 3$ counting qubits and 4 work qubits. The multiplications by 7 and $4 = 7^2 \bmod 15$ are compiled into controlled SWAPs (each CNOT–Toffoli–CNOT, Toffoli = 6 CNOTs) and CNOTs; the inverse QFT uses 3 CNOTs (one SWAP) and 3 controlled phases. In total the circuit has 127 CNOTs. It was simulated with the CUDA-Q `density-matrix-cpu` target and a depolarizing noise model: a one-qubit channel with parameter p after every H, R_z and X, and a two-qubit channel with parameter p after every CNOT and controlled phase. The probability that one shot yields the factors 3×5 was (re-executed, exact from the density matrix):

p	0	0.001	0.003	0.01	0.03
success probability	0.750	0.740	0.726	0.703	0.665

If the counting register were completely random (uniform y), the success probability would be 0.375. These are simulator results, not measurements on a quantum processor.

Q.79 PARAGRAPH BASED · Error-free probability vs success

At $p = 0.01$, the probability that *all* 127 CNOTs are free of error is about 0.99^{127}. Compare it with the success probability.

(A) $0.99^{127} \approx 0.88$, equal to the success probability

(B) ✓ $0.99^{127} \approx 0.28$, yet the success probability is 0.703: many errors do not change y or still lead to a correct order

(C) $0.99^{127} \approx 0.01$, so the result is pure noise

(D) $0.99^{127} \approx 0.28$, so the success probability must be at most 0.28

Answer: (B) Detailed solution

Correct answer. **(B)** $0.99^{127} \approx 0.28$, yet the success probability is 0.703: many errors do not change y or still lead to a correct order

Concepts required. The error-free probability bounds nothing: many errors are harmless for the final answer.

Relevant formulae. $0.99^{127} = e^{-1.276} = 0.279$; measured (simulated) success 0.703.

Step-by-step explanation. $0.99^{127} = e^{127 \ln 0.99} = e^{-1.276} = 0.279$. The success probability is not bounded by the error-free probability: errors on work qubits after their last use, phase errors that commute with measurement, or errors that shift y to another successful value do not cause failure. Also, a random y still succeeds with probability 0.375.

Why the other options are incorrect.

(A) $0.99^{127} \approx 0.28$, not 0.88; $0.88 \approx 0.99^{13}$.

(C) $0.99^{127} \approx 0.28$, not 0.01; and 0.703 is well above the random baseline 0.375.

(D) Errors that do not change y, or move it to another successful value, do not cause failure; $0.703 > 0.28$ contradicts this option.

Key takeaway. Estimate error impact by simulation, not by the naive product of gate fidelities.

Q.80 PARAGRAPH BASED · Interpreting noisy results

Which conclusion is supported by the table?

(A) the results show that a quantum processor factored 15

(B) noise increases the success probability

(C) at $p = 0.03$ the circuit is completely random

(D) ✓ the success probability decreases monotonically from 0.750 towards the random baseline 0.375; at $p = 0.03$ it is 0.665, still above the baseline

Answer: (D) Detailed solution

Correct answer. (D) the success probability decreases monotonically from 0.750 towards the random baseline 0.375; at $p = 0.03$ it is 0.665, still above the baseline

Concepts required. Noise interpolates between the ideal distribution and the uniform baseline.

Relevant formulae. Ideal 0.750; baseline 0.375; $p = 0.03$: 0.665.

Step-by-step explanation. The values fall $0.750 \to 0.740 \to 0.726 \to 0.703 \to 0.665$; the gap to the uniform baseline (0.375) shrinks but has not vanished. These are density-matrix *simulations* with an idealised noise model — no quantum processor was used, and factoring 15 with a compiled circuit that uses the known order is a demonstration, not evidence of quantum advantage.

Why the other options are incorrect.

(A) These are simulations on `density-matrix-cpu`; no quantum processor was used.

(B) The values decrease with p; noise never helped here.

(C) 0.665 is far above the random baseline 0.375; the circuit is not yet completely random.

Key takeaway. Label simulated results as simulations; compare them with the right baseline.

SILICO FELLER